arXiv:2608.16466v2 [stat.ME] 23 Aug 2026

Deep adaptive design with an evidential bias criterion

David Chen* Michael Evans† Xinwei Li ‡ Prateek Bansal§
David J. Nott¶

Abstract

Bayesian optimal experimental design (BOED) aims to collect informative data by optimizing an expected utility reflecting the goals of an experiment. However, this optimization is computationally challenging for common utilities and complex models. This is especially so for sequential or adaptive designs, where design and data collection alternate, so that feedback from already observed data must be taken into account. Most existing BOED research employs information gain as the utility, leading to the expected information gain (EIG) criterion. While EIG is widely useful, it may not always adequately reflect experimental goals. EIG can be viewed as rewarding experiments that produce large positive evidence for the truth on average, but it does not directly control the risk of an experiment producing misleading evidence. Here we consider an alternative criterion, which we call bias against (BA), that prioritizes such control. To address computational challenges when applying this criterion for adaptive design, we consider a policy-based deep adaptive design framework, which has previously been used for the EIG criterion. Minimizing a tractable upper bound on the BA objective is equivalent to maximizing a variance-penalized EIG criterion, and we optimize the latter by approximating it by Monte Carlo and learning design policies using stochastic gradient methods. The differences between BA and EIG designs are demonstrated in several examples including the adaptive design of a complex discrete choice experiment.

Keywords: Bias against; Deep adaptive design; Expected information gain; Optimal Bayesian experimental design; Sequential experimental design.

*Department of Statistics and Data Science, National University of Singapore.
†Department of Statistical Sciences, University of Toronto, Toronto, ON M5S 3G3, Canada.
‡Department of Civil and Environmental Engineering, National University of Singappore.
§Department of Civil and Environmental Engineering, National University of Singapore and Department of Statistics and Data Science, National University of SIngapore.
¶Corresponding author: standj@nus.edu.sg. Department of Statistics and Data Science, National University of Singapore and Institute of Operations Research and Analytics, National University of Singapore.

1 Introduction

Collecting the most informative data possible is crucial to researchers in many different fields, particularly when data collection is expensive. Bayesian optimal experimental design (BOED) proposes to choose the best design by maximizing an expected utility. Popular design criteria are expectations, under the joint Bayesian model, of nonlinear functions of the posterior density or marginal likelihood, quantities that are intractable in most problems. Because the intractabilities appear as an inner expectation inside a nonlinearity, the criterion cannot be written as a single expectation, and direct approximation requires nested Monte Carlo methods (Rainforth et al., 2018). This is referred to as the "double intractability" of the design objective. Sequential or adaptive design problems which alternate between observing data and choosing design variables are particularly difficult, due to the need to incorporate the feedback from past data. Recent progress in addressing the computational burden has used policy-based approaches, where a policy function specifying the design choice as a function of past data is optimized using techniques inspired by dynamic programming, reinforcement learning and deep learning (Huan and Marzouk, 2016; Foster et al., 2021; Ivanova et al., 2021; Shen and Huan, 2023; Blau et al., 2022, 2023; Lim et al., 2022). These methods learn better designs than traditional greedy approaches and can be amortized, which means that after learning the policy the computation of designs can be done very rapidly.

Existing work on amortized adaptive design has focused mostly on using the expected information gain (EIG) criterion (Lindley, 1956). While EIG is widely used, alternative criteria can be preferable in some situations, and here we explore an evidential bias criterion and the construction of designs using the deep adaptive design approach of Foster et al. (2021). EIG can be thought of as rewarding large positive evidence for the truth on average, whereas the evidential bias criterion considered here is a probability of obtaining misleading evidence against the truth. The concern with controlling misleading evidence in design is not new. For example, Royall (1997) argues that likelihood ratios are an appropriate way to quantify the strength of statistical evidence, and Royall (2000) studies the probability that a likelihood ratio favours a false hypothesis, with implications for design questions such as the choice of sample size. Blume (2002) extends the discussion to a range of study designs. The criterion we consider here addresses the same concern from a Bayesian perspective. It is built from the relative belief ratio, which measures the change in belief from prior to posterior, and the probability of misleading evidence is computed under the joint Bayesian model. Our concern in this work is with making such a criterion usable for amortized sequential design. This paper makes four contributions. First, we propose the use of evidential bias – also called the bias

against (BA) in Evans (2015) – as a useful criterion in sequential design problems. Secondly, we develop an upper bound on the criterion that is insightful and easier to optimize than the original criterion using stochastic gradient methods. Optimizing the upper bound is equivalent to maximizing a variance penalized version of the EIG criterion, and is related to the risk-aware BOED approach of Shen and Huan (2026). Thirdly, we adapt the deep adaptive design framework of Foster et al. (2021) to computation of designs for the BA criterion. Finally, we demonstrate how designs based on bias against differ from EIG-based designs in several examples, including the adaptive design of a complex discrete choice experiment.

Our contributions build on recent literature on computational methods for Bayesian sequential design. An important early work by Huan and Marzouk (2016) computed non-myopic Bayesian adaptive designs by optimizing a policy for a finite-horizon Markov decision process. They use an implicit representation of the optimal policy and solve for it using approximate dynamic programming. Foster et al. (2021) make the design policy explicit, and extend the contrastive bounds of Foster et al. (2020) to the sequential setting. The bounds are easily estimated by Monte Carlo and optimized using stochastic gradient approaches. Their deep adaptive design (DAD) approach has been generalized in several directions, such as for implicit likelihoods (Ivanova et al., 2021; Lim et al., 2022) and semi-amortized algorithms (Ivanova et al., 2024). Blau et al. (2022) combine a reinforcement learning formulation of Bayesian adaptive design with the variational bounds in Foster et al. (2021) and use well-developed policy optimization methods from reinforcement learning to obtain better designs, including for discrete design problems. Shen and Huan (2023) also consider reinforcement learning approaches, employing an actor-critic policy gradient method for the optimization. Shen et al. (2025) extend Shen and Huan (2023) by employing variational "one point" reward formulations and incorporating model uncertainty, nuisance parameters, handling of implicit models and more flexible design objectives. Blau et al. (2023) address the difficulty of estimating the bounds of Foster et al. (2021) when the EIG is large, which they do by proposing a cross-entropy estimator based on a sequential version of the Barber-Agakov bound (Barber and Agakov, 2003). A related approach is that of Bracher et al. (2026), where the authors avoid computation of the approximate posterior density and use diffusion approaches. Phillips et al. (2026) obtain singly intractable objectives using score-matching methods for optimizing the EIG criterion.

The above works focus mostly on EIG as the design criterion, although some works such as Shen et al. (2025) consider EIG for subsets of parameters, model indicators or predictive quantities of interest. Huang et al. (2026) consider similar design objectives which can be dynamically chosen, with amortization over the choice of design and the posterior estimation.

Many alternatives to EIG have been explored in BOED, but little of the existing work has been in the context of amortized sequential design. Some recent exceptions are Huang et al. (2024), who consider design using predictive utilities which reflect downstream decision-making tasks, and Rossa et al. (2026), who formulate Bayesian design decision-theoretically in terms of expected future loss, leading to singly intractable objectives. Their approach requires two policies to be learnt, which they call the design and action policies.

One goal of the present paper is to produce adaptive designs using an alternative to EIG, which is focused on the avoidance of misleading evidence. The existing work most closely related to ours is Shen and Huan (2026), where the authors consider design criteria that penalize expected utility with a utility variance. They focus mostly on the prior-to-posterior Kullback-Leibler divergence as the utility, leading to a penalized EIG criterion. Penalizing an expected utility by its variance is one of many ways of expressing risk aversion in design, and Huan et al. (2024, Section 6.2) discuss some of the alternatives, including quantile-based measures such as the value-at-risk and conditional value-at-risk (Rockafellar and Uryasev, 2000), worst-case criteria and entropic risk. They observe that these approaches have so far seen relatively little use in nonlinear design with general utilities. We consider penalized EIG criteria in our work, although we use a different variance penalty term to Shen and Huan (2026), and focus on adaptive designs using policy learning. Our approach also has a different motivation in terms of avoiding evidential bias, an interpretation that is helpful in choosing the penalty parameter. The connections between our approach and that of Shen and Huan (2026) are discussed in detail in Section 3.

In the next section we discuss the Bayesian formulation of optimal experimental design in batch and sequential settings. We then explain the EIG criterion and the alternative BA criterion that we focus on in our work. In Section 3 we derive an upper bound on the BA criterion using the Paley-Zygmund inequality (Paley and Zygmund, 1932). The bound has a simple expression in terms of the mean and variance of the log relative belief ratio, and we show that minimizing it is equivalent to maximizing a variance penalized EIG criterion, the latter being more convenient to optimize. We also compare the bias against, its upper bound and EIG theoretically. Section 4 discusses computational issues, while Section 5 considers three examples, including the adaptive design of a complex discrete choice experiment.

2 Bayesian optimal experimental design

First we briefly review basic ideas of BOED. For further background, Huan et al. (2024) give a comprehensive recent review, and Rainforth et al. (2024) and Ryan et al. (2016) discuss

computational aspects. Chaloner and Verdinelli (1995) consider classical design problems from the point of view of Bayesian decision theory.

2.1 Batch and sequential design

We introduce some notation first. Denote data to be collected as $y = (y_1, \ldots, y_T)$ where $y_t \in \mathcal{Y}_t$, $t = 1, \ldots, T$. There is an assumed parametric model for y_t having parameters θ, with a density $p(y_t|x_t, \theta)$, where $x_t \in \mathcal{X}_t$ is some design variable to be chosen. The prior density for θ is denoted $p(\theta)$. Write $x = (x_1, \ldots, x_T)$, and the model for y given x, θ is

$$p(y|x, \theta) = \prod_{t=1}^{T} p(y_t|x_t, \theta).$$

We think of $t = 1, \ldots, T$ as indexing a sequence of experiments, and x_t is chosen before y_t is observed. The information available for choosing x_t comes from past experiments, $\mathcal{I}_{t-1} = \{(y_j, x_j) : j < t\} \in \mathcal{H}_t$ with $\mathcal{I}_0$ being the empty set.

In traditional "batch" experimental design we consider $T = 1$ so that there is a single experiment where the whole design x is chosen before observing y. A utility function $U(y, x, \theta)$ is considered, and we choose x by maximizing expected utility:

$$x^* := \arg\max_x E_{y\sim p(y|x)} \left(E_{\theta\sim p(\theta|y,x)} \left(U(y, x, \theta) \right) \right),$$

where

$$p(\theta|y, x) \propto p(\theta) \prod_{t=1}^{T} p(y_t|x_t, \theta),$$

is the posterior density for θ, and $p(y|x) = E_{\theta\sim p(\theta)}(p(y|x, \theta))$. Equivalently, we can write

$$x^* = \arg\max_x E_{\theta\sim p(\theta)} \left(E_{y\sim p(y|x,\theta)} \left(U(y, x, \theta) \right) \right).$$

Throughout the manuscript, we will sometimes write the posterior density $p(\theta|y, x)$ as $p(\theta|y)$ for simpler notation where no confusion will arise in suppressing dependence on x.

In contrast to batch experimental design, our work will focus on adaptive experimental design (also called sequential experimental design) where $T > 1$ and at step t the choice of x_t can depend on $\mathcal{I}_{t-1}$, so that feedback from previoius observations must be accounted for in design decisions. Traditional (greedy) Bayesian adaptive design would make the choice of x_t by maximizing the expected value of a utility $U_t(y_t, x_t, \mathcal{I}_{t-1}, \theta)$:

$$x_t^* = \arg\max_{x_t} E_{y_t\sim p(y_t|\mathcal{I}_{t-1},x_t)} \left(E_{\theta\sim p(\theta|\mathcal{I}_t)} \left(U_t(y_t, x_t, \mathcal{I}_{t-1}, \theta) \right) \right),$$

where $p(y_t|\mathcal{I}_{t-1}, x_t) = E_{\theta\sim p(\theta|\mathcal{I}_{t-1})}(p(y_t|\theta, x_t))$ and $p(\theta|\mathcal{I}_t)$ is the posterior density of θ given $\mathcal{I}_t$. Equivalently, we can write

$$x_t^* = \arg\max_{x_t} E_{\theta\sim p(\theta|\mathcal{I}_{t-1})}\left(E_{y_t\sim p(y_t|x_t,\theta)}\left(U_t(y_t, x_t, \mathcal{I}_{t-1}, \theta)\right)\right).$$

This traditional adaptive BOED approach has two main problems. Firstly, computation is difficult because we need to repeatedly approximate the posterior distribution after each experiment. Secondly, because we aim to only optimize the expected utility for the next experiment, we ignore the effects of future data. This leads to so-called "greedy" or "myopic" designs. Amortized policy-based designs (e.g. Rainforth et al. 2024) can address both of these issues.

2.2 Amortized sequential design

Policy-based sequential design methods have become quite popular recently, and these approaches were first used for adaptive BOED by Huan and Marzouk (2016) to the best of our knowledge. In their formulation the specification of the policy is implicit, and here we will follow Foster et al. (2021) who use an explicit policy function depending on learnable parameters φ, defining for any fixed step t a mapping $\eta_\varphi : \mathcal{H}_t \to \mathcal{X}_t$. The construction of this mapping is important and should respect the symmetries of the problem. We will discuss this later. Foster et al. (2021) learn the parameters φ to achieve the best global value of the design criterion over the entire sequence of T data-collection steps. This procedure does not require approximating a posterior density at any stage, and, as noted above, it learns better designs by optimizing the global design criterion directly rather than greedily optimizing local criteria, which may yield inferior solutions for the global objective. A further benefit is that, once the policy is learned, generating designs is very fast which is an important consideration in many applications.

2.3 Expected information gain and relative belief

Foster et al. (2021) consider deep adaptive design for the expected information gain criterion, denoted here as $\mathrm{EIG}(x)$. We consider the batch (non-adaptive) case first. There are many equivalent formulations of EIG, and we begin with one expressed in terms of the prior-to-posterior Kullback-Leibler (KL) divergence (Kullback and Leibler, 1951). For two distributions Q and $\widetilde{Q}$ with densities q and $\widetilde{q}$ respectively, the KL divergence of Q from $\widetilde{Q}$ is

$$\mathrm{KL}(\widetilde{q}(\theta) \,\|\, q(\theta)) := E_{\theta\sim\widetilde{q}(\theta)}(\log\widetilde{q}(\theta) - \log q(\theta)),$$

which measures how far Q is from $\widetilde{Q}$ and is zero if and only if $\widetilde{Q} = Q$.

The EIG can be defined as

$$\begin{aligned} \mathrm{EIG}(x) &:= E_{y\sim p(y|x)} E_{\theta\sim p(\theta|y,x)} \left\{ \log \frac{p(\theta|y,x)}{p(\theta)} \right\} \\ &= E_{y\sim p(y|x)} \left\{ \mathrm{KL}(p(\theta|y,x) \,\|\, p(\theta)) \right\}, \end{aligned} \tag{1}$$

so that $\mathrm{EIG}(x)$ measures how far the prior is from the posterior on average for the design x. If $\mathrm{EIG}(x)$ is large this is good, since a large change in the Bayesian update on average indicates that the experiment is highly informative. We can immediately notice the potential difficulty in optimizing $\mathrm{EIG}(x)$. In the KL divergence, evaluating $p(\theta|y,x)$ pointwise involves knowing the normalizing constant $p(y|x) = E_{\theta\sim p(\theta)}(p(y|x,\theta))$, which is an intractable integral in the form of an expectation that appears inside a log term. Optimizing the nested expectation, which does not collapse to a single joint expectation, is challenging.

An equivalent definition of $\mathrm{EIG}(x)$ that follows from (1) is

$$\mathrm{EIG}(x) = E_{y\sim p(y|x)}(H(\theta) - H(\theta|y)), \tag{2}$$

where

$$H(\theta) = -E_{\theta\sim p(\theta)}(\log p(\theta)) \;\text{ and }\; H(\theta|y) = -E_{\theta\sim p(\theta|y)}(\log p(\theta|y)),$$

which is the expected reduction in Shannon or differential entropy between prior and posterior for the experiment x. The quantity $H(\theta) - H(\theta|y)$ is called the information gain, and its use as a utility for Bayesian experimental design was first suggested in Lindley (1956).

The ratio that appears in (1) is called the relative belief ratio (Evans, 2015), and we will introduce some special notation for it,

$$\mathrm{RB}(\theta; y, x) := \frac{p(\theta|y,x)}{p(\theta)}. \tag{3}$$

The bias against criterion that is introduced in the next section is defined in terms of the relative belief ratio. The meaning of $\mathrm{RB}(\theta; y, x)$ is the following: if $\mathrm{RB}(\theta; y, x) > 1$ for a given θ, then the data y gives evidence in favour of θ being the true value. This is because $\mathrm{RB}(\theta; y, x) > 1$ implies that the posterior is larger than the prior at θ, so our belief that θ is the true value has increased. Similarly, if $\mathrm{RB}(\theta; y, x) < 1$, there is evidence against θ being the true value, whereas if $\mathrm{RB}(\theta; y, x) = 1$ then there is no evidence either way. We will call the set of parameters with evidence in favour,

$$\left\{ \theta : \frac{p(\theta|y,x)}{p(\theta)} > 1 \right\}, \tag{4}$$

the plausible region. Introducing the notation

$$Z(y, x, \theta) := \log \mathrm{RB}(\theta; y, x), \tag{5}$$

we see from (1) that $\mathrm{EIG}(x) = E_{y\sim p(y|x),\theta\sim p(\theta|y,x)}(Z(y, x, \theta))$, the expected log relative belief.

It is useful to have an alternative expression for the relative belief. By rearranging Bayes' rule,

$$\mathrm{RB}(\theta; y, x) = \frac{p(\theta|y, x)}{p(\theta)} = \frac{p(y|x, \theta)}{p(y|x)} \tag{6}$$

and then using this alternative expression in (1) we obtain a corresponding alternative expression for $\mathrm{EIG}(x)$,

$$\mathrm{EIG}(x) = E_{\theta\sim p(\theta)} E_{y\sim p(y|\theta)} \left\{ \log \frac{p(y|x, \theta)}{p(y|x)} \right\}. \tag{7}$$

For design, we want to maximize $\mathrm{EIG}(x)$. Foster et al. (2020) consider independent draws $\theta_0, \theta_1, \ldots, \theta_L \sim p(\theta)$ and estimate $p(y|x)$ in (7) by either

$$\frac{1}{L+1} \sum_{l=0}^{L} p(y|x, \theta_l)$$

or

$$\frac{1}{L} \sum_{l=1}^{L} p(y|x, \theta_l).$$

This results in two approximations to the EIG criterion,

$$\mathrm{EIG}_l(x) = E_{\theta_0,\theta_1,\cdots,\theta_L\sim p(\theta)} E_{y\sim p(y|\theta_0)} \left\{ \log \frac{p(y|x, \theta_0)}{\frac{1}{L+1}\sum_{l=0}^{L} p(y|x, \theta_l)} \right\}, \tag{8}$$

and

$$\mathrm{EIG}_u(x) = E_{\theta_0,\theta_1,\cdots,\theta_L\sim p(\theta)} E_{y\sim p(y|\theta_0)} \left\{ \log \frac{p(y|x, \theta_0)}{\frac{1}{L}\sum_{l=1}^{L} p(y|x, \theta_l)} \right\}. \tag{9}$$

$\mathrm{EIG}_l(x)$ and $\mathrm{EIG}_u(x)$ are respectively lower and upper bounds for (7). The lower bound is the prior contrastive estimation (PCE) bound of Foster et al. (2020), and the upper bound is the standard nested Monte Carlo estimator, whose upward bias follows from Jensen's inequality. See Rainforth et al. (2018) for the convergence properties of such estimators. $\mathrm{EIG}_l(x)$ and $\mathrm{EIG}_u(x)$ are sometimes called contrastive bounds, because of the use of contrastive prior samples $\theta_1, \ldots, \theta_L$. Foster et al. (2021) extend them to the sequential setting.

The bounds become tight as $L \to \infty$, and they can be optimized using stochastic gradient ascent. For optimizing φ with a policy mapping in the adaptive design context, Foster et al.

(2021) use the lower bound for reasons of numerical stability. In this case, since x_t is determined from φ given $\mathcal{I}_{t-1}$ for all t, the design criteria (8) and (9) are functions of φ. To make this clearer, instead of writing $p(y|x,\theta)$ we write

$$p(y|\varphi,\theta) = \prod_{t=1}^{T} p(y_t|x_t = \eta_\varphi(\mathcal{I}_{t-1}),\theta),$$

and $p(y|\varphi) = E_{\theta\sim p(\theta)}(p(y|\theta,\varphi))$. With this notation, we write (8) as

$$\mathrm{EIG}_l(\varphi) = E_{\theta,\theta_1,\cdots,\theta_L\sim p(\theta)} E_{y\sim p(y|\varphi,\theta)} \left\{ \log \frac{p(y|\varphi,\theta)}{\frac{1}{L+1}\sum_{l=0}^{L} p(y|\varphi,\theta_l)} \right\}. \tag{10}$$

and it is this criterion that Foster et al. (2021) optimize to learn the policy parameters φ.

3 Bias against criterion and its upper bound

As an alternative to EIG, we now consider the bias against criterion for design, the main object of study for the rest of the paper. Again, we begin with the batch case. For a given parameter value θ, the bias against θ is

$$\mathrm{BA}(\theta;x) := P_{y\sim p(y|x,\theta)} \left(\frac{p(y|\theta,x)}{p(y|x)} \leq 1 \right), \tag{11}$$

where $p(y|x) = E_{\theta\sim p(\theta)}(p(y|x,\theta))$. For a given θ, $\mathrm{BA}(\theta;x)$ is the probability that the true value of θ does not belong to the plausible region for data y generated at θ using design x. We write $\mathrm{BA}(x)$ (which we call simply the bias against) for its expectation,

$$\mathrm{BA}(x) := E_{\theta\sim p(\theta)} \left(\mathrm{BA}(\theta;x)\right) = P_{\theta\sim p(\theta), y\sim p(y|x,\theta)} \left(\frac{p(y|\theta,x)}{p(y|x)} \leq 1 \right). \tag{12}$$

$\mathrm{BA}(x)$ is the average of $\mathrm{BA}(\theta;x)$ over the prior for θ, and is the probability under the joint Bayesian model that misleading evidence is produced using design x. The discussion above generalizes to the case of a parameter of interest that is not the full parameter θ, but we don't consider this here for simplicity. We can rewrite (12) as

$$\mathrm{BA}(x) := E_{y\sim p(y|x)} \left\{ P_{\theta\sim p(\theta|y,x)} \left(\frac{p(y|\theta,x)}{p(y|x)} \leq 1 \right) \right\}. \tag{13}$$

which shows that minimizing $\mathrm{BA}(x)$ corresponds to maximizing expected utility for the utility function

$$-P_{\theta\sim p(\theta|y,x)} \left(\frac{p(y|\theta,x)}{p(y|x)} \leq 1 \right).$$

Maximizing the expected value of this utility means having a small probability that the experiment using x will produce misleading evidence.

Direct stochastic optimization of $\mathrm{BA}(x)$ is difficult: it is a probability, for which the gradients are uninformative when the design is already reasonably good. We therefore work with an upper bound on $\mathrm{BA}(x)$ which is easier to optimize. The bound is obtained from the Paley-Zygmund inequality (Paley and Zygmund, 1932) and has an interpretation in terms of the mean and variance of the log relative belief ratio. We first state the relevant inequality, then derive the bound, and finally describe computation in the policy-based setting.

3.1 The Paley-Zygmund inequality and a coefficient-of-variation bound

Recall the notation $\mathrm{RB}(\theta; y, x) := p(y|x,\theta)/p(y|x)$ for the relative belief ratio at θ when the data are y and the design is x, and $Z(y,x,\theta) := \log \mathrm{RB}(\theta; y, x) = \log p(y|x,\theta) - \log p(y|x)$ for its logarithm. Throughout this section, expectations are taken under the joint Bayesian model $\theta \sim p(\theta)$, $y \sim p(y|x,\theta)$, and we write $E(\cdot)$ and $\mathrm{Var}(\cdot)$ for the joint expectation and variance. With this notation, $\mathrm{EIG}(x) = E(Z(y,x,\theta))$ and

$$\mathrm{BA}(x) = P(Z(y,x,\theta) \le 0).$$

Our bound is derived from the following modified form of the Paley-Zygmund inequality. It is usually stated for a non-negative random variable, but the proof only requires the mean to be non-negative.

Proposition 1. *Let Z be a random variable with $0 < \mathrm{Var}(Z) < \infty$ and $E(Z) \ge 0$. Then for all $\delta \in [0,1]$,*

$$P(Z > \delta E(Z)) \ge \frac{(1-\delta)^2 E(Z)^2}{E((Z - \delta E(Z))^2)} = \frac{(1-\delta)^2 E(Z)^2}{\mathrm{Var}(Z) + (1-\delta)^2 E(Z)^2} \tag{14}$$

$$P(Z \le \delta E(Z)) \le \frac{E((Z - E(Z))^2)}{E((Z - \delta E(Z))^2)} = \frac{\mathrm{Var}(Z)}{\mathrm{Var}(Z) + (1-\delta)^2 E(Z)^2}. \tag{15}$$

The proof is an easy application of the Cauchy-Schwarz inequality, given in Appendix A. Applying Proposition 1 to $Z = Z(y,x,\theta)$ with $\delta = 0$, and noting that $E(Z(y,x,\theta)) = \mathrm{EIG}(x) \ge 0$, we obtain the following.

Corollary 1. *If $0 < \mathrm{Var}(Z(y,x,\theta)) < \infty$, then*

$$\mathrm{BA}(x) \le \frac{\mathrm{Var}(Z(y,x,\theta))}{\mathrm{Var}(Z(y,x,\theta)) + \mathrm{EIG}(x)^2} =: \widetilde{\mathrm{BA}}(x). \tag{16}$$

The quantity $\widetilde{\mathrm{BA}}(x)$ defined in (16) is the upper bound on $\mathrm{BA}(x)$ that we use throughout the remainder of the paper. Rearranging (16) gives

$$\frac{\mathrm{sd}(Z(y,x,\theta))}{\mathrm{EIG}(x)} \geq \sqrt{\frac{\mathrm{BA}(x)}{1-\mathrm{BA}(x)}}, \tag{17}$$

where $\mathrm{sd}(Z(y,x,\theta)) = \sqrt{\mathrm{Var}(Z(y,x,\theta))}$ is the standard deviation of $Z(y,x,\theta)$, and the left-hand side of (17) is the coefficient of variation of the log relative belief ratio under the joint Bayesian model. This is a unitless measure of how reliably the experiment produces evidence in favour of the true parameter value. The inequality (17) implies that if the coefficient of variation of the log relative belief ratio is small, then the bias against is also small.

3.2 A variance-penalized EIG criterion

To understand better what minimizing $\widetilde{\mathrm{BA}}(x)$ does, divide numerator and denominator on the right-hand side of (16) by $\mathrm{Var}(Z(y,x,\theta))$ (assuming this is non-zero) to obtain

$$\widetilde{\mathrm{BA}}(x) = \frac{1}{1+\mathrm{EIG}(x)^2/\mathrm{Var}(Z(y,x,\theta))}. \tag{18}$$

Since $\mathrm{EIG}(x) = E(Z(y,x,\theta)) \geq 0$, this shows that minimizing $\widetilde{\mathrm{BA}}(x)$ is equivalent to maximizing

$$R(x) := \frac{\mathrm{EIG}(x)}{\mathrm{sd}(Z(y,x,\theta))} = \frac{E(Z(y,x,\theta))}{\sqrt{\mathrm{Var}(Z(y,x,\theta))}}, \tag{19}$$

which is the inverse of the coefficient of variation appearing in (17). The numerator is the expected information gain, which rewards designs that produce large evidence for the true value on average. The denominator $\mathrm{sd}(Z(y,x,\theta))$ penalizes variability of the log relative belief ratio, expressing the preference for reliable evidence. Maximizing EIG while ignoring its variability, which is what the standard EIG criterion does, may produce designs for which evidence is large on average but highly variable. Optimizing for average behaviour, however, may not effectively control the risk of observing misleading evidence. Maximizing $R(x)$ is equivalent to maximizing $\mathrm{EIG}(x)$ if the log relative belief ratio has a variance independent of the design, but this is not the case in general.

Maximizing $R(x)$ directly is awkward because estimating the ratio using Monte Carlo can be unstable. However, any maximizer x^* of $R(x)$ maximizes $\mathrm{EIG}(x)$ subject to a constraint on $\mathrm{Var}(Z(y,x,\theta))$. A Lagrangian argument (see Appendix B) shows that x^* is also a maximizer of

$$L_\alpha(x) := E(Z(y,x,\theta)) - \alpha\,\mathrm{Var}(Z(y,x,\theta)), \tag{20}$$

for some value $\alpha \geq 0$. Using a penalized objective such as (20) to optimize a ratio is a standard technique in fractional programming (Dinkelbach, 1967). Dinkelbach's algorithm alternates between optimizing x for fixed α and optimizing α for fixed x. We do not do this here, since, as we explain below, the whole path of designs traced out by varying α is of interest, not just the solution optimising $R(x)$. We therefore optimize $L_\alpha(x)$ over a grid of α values, and select the value from the grid maximizing the original ratio $R(x)$ only if we wish to choose a single α.

The penalized criterion (20) is highly interpretable. A similar criterion is considered by Shen and Huan (2026). At $\alpha = 0$ (20) reduces to EIG, and increasing α trades expected information gain (the expected log relative belief) for reduced variability of the log relative belief. The criterion of Shen and Huan (2026) is not exactly the same as (20), as they consider a variance penalty of the form (in our notation) $\mathrm{Var}_{y\sim p(y|x)}(\mathrm{KL}(p(\theta|y,x) \,\|\, p(\theta)))$, which is the variance of a prior to posterior KL divergence utility with respect to $p(y|x)$. Shen and Huan (2026) consider batch design, with Shen (2023, Chapter 5) discussing the adaptive case. Further discussion of connections with their approach is given in the next subsection.

We can give an interesting interpretation of the entire path of solutions for the design as α varies in (20), in terms of an extended notion of misleading evidence. Fix a constant $t \geq 0$ and consider the event $\{Z(y,x,\theta) \leq t\}$, that the log relative belief ratio fails to exceed t. The choice $t = 0$ recovers the bias against. Applying Proposition 1 to $Z = Z(y,x,\theta)$ with $\delta = t/\mathrm{EIG}(x)$, which lies in $[0,1]$ when $0 \leq t \leq \mathrm{EIG}(x)$, gives

$$P(Z(y,x,\theta) \leq t) \leq \frac{\mathrm{Var}(Z(y,x,\theta))}{\mathrm{Var}(Z(y,x,\theta)) + (\mathrm{EIG}(x) - t)^2}. \tag{21}$$

Minimizing the bound (21) is equivalent to maximizing

$$R_t(x) := \frac{\mathrm{EIG}(x) - t}{\mathrm{sd}(Z(y,x,\theta))}, \tag{22}$$

which differs from (19) by replacing $\mathrm{EIG}(x)$ with the offset $\mathrm{EIG}(x) - t$. The optimizer of (22) is also an optimizer of the penalized criterion

$$L_{\alpha,t}(x) = E(Z(y,x,\theta)) - t - \alpha\,\mathrm{Var}(Z(y,x,\theta))$$

for some α, and the family of designs traced out by optimizing this criterion is the same as the family of solutions traced out by varying α in $L_\alpha(x)$. However if α in (20) is chosen to tighten the bound (21) as much as possible, the design which is chosen varies with t. So different choices of α correspond to optimizing a bound for $P(Z(y,x,\theta) \leq t)$ for different choices of t.

Proposition 2 in Appendix B links maximizers of $R_t(x)$ to maximizers of $L_\alpha(x)$. Although we assumed $\mathrm{EIG}(x) > t$ in application of the Paley-Zygmund inequality, it is not necessary to

restrict the domain of optimization using this condition. Provided the design class contains at least one x with $\mathrm{EIG}(x) \geq t$, so that $\max_x R_t(x) \geq 0$, any design with $\mathrm{EIG}(x) < t$ has $R_t(x) < 0$ and cannot be a maximizer. At an interior maximizer x^* the corresponding penalty is

$$\alpha^*(t) = \frac{\mathrm{EIG}(x^*) - t}{2\,\mathrm{Var}(Z(y, x^*, \theta))}, \tag{23}$$

obtained by applying Proposition 2 with offset t, where $x^* = x^*(t)$ is the maximizer of R_t. Writing $r^*(t) := \max_x R_t(x)$, and since $R_t(x) = (\mathrm{EIG}(x) - t)/\sqrt{\mathrm{Var}(Z(y, x, \theta))}$ is decreasing in t for fixed x, it follows that $r^*(t)$ is non-increasing in t. Choosing α in (20) corresponds to choosing a threshold t for the misleading-evidence event $\{Z(y, x, \theta) \leq t\}$ whose bound (21) is being tightened. In practice (23) cannot be set in advance, as it depends on the mean and variance of Z at the optimum. For the computations in Section 4, we search over a grid for α, but the above discussion clarifies that each penalty value corresponds to tightening the bound on a particular threshold event.

3.3 Relationship to risk-aware BOED

The discussion of the previous subsection leads to connections between our approach and the risk aware BOED method of Shen and Huan (2026). It is possible to justify their criteria as approximate expected utility maximization using an upper bound based on the Paley-Zygmund inequality. To see this, suppose that $U(y, x)$ is a utility function, with $E(U(y, x)) \geq 0$ and $0 < \mathrm{Var}(U(y, x)) < \infty$ with $y \sim p(y|x)$. Shen and Huan (2026) focus mostly on the choice $U(y, x) = \mathrm{KL}(p(\theta|y) \,\|\, p(\theta))$ in their work, and this utility leads to the penalized EIG criterion. Now consider obtaining a design by minimizing $P(U(y, x) \leq t)$ for $y \sim p(y|x)$, corresponding to maximizing a utility of $-P(U(y, x) \leq t)$. For $0 \leq t \leq E(U(y, x))$, applying the Paley-Zygmund inequality similar to before, gives an upper bound

$$P(U(y, x) \leq t) \leq \frac{\mathrm{Var}(U(y, x))}{\mathrm{Var}(U(y, x)) + (E(U(y, x)) - t)^2},$$

and minimizing this is equivalent to maximizing

$$\frac{E(U(y, x)) - t}{\sqrt{\mathrm{Var}(U(y, x))}}.$$

An optimum of this upper bound is an optimum of the penalized criterion

$$E(U(y, x)) - t - \alpha \mathrm{Var}(U(y, x)),$$

for some value of the penalty $\alpha > 0$, where the family of designs traced out does not depend on t. In our bias against approach the choice of threshold $t = 0$ for tightening the bound is

natural if we wish to choose a single α. This is because $t = 0$ corresponds to a log relative belief of 0, or a relative belief of 1, which has a special meaning as the boundary between evidence in favour and evidence against. However, for the Kullback-Leibler utility and most other choices, there might be no obvious choice of threshold t to use if we choose α by tightening the bound.

3.4 Theoretical comparisons of EIG and R

Now that we have discussed the motivation for the BA criterion and its upper bound, we give some theoretical comparisons with EIG. We start with a simple example that is informative about the relationship between EIG(x), BA(x) and $R(x)$. In the example we take x as fixed, although the sample size could be taken as the design variable. Since there is no varying x, we write EIG, BA and R for the EIG criterion, bias against and ratio criterion. We have observed earlier that the BA upper bound $\widetilde{\text{BA}}$ is a decreasing function of R. Consequently, by making R large enough, the bias against can be made as close to zero as we like. No such guarantee holds for the EIG criterion, and our example demonstrates this. It shows a situation where EIG $\to \infty$ but BA $\to q > 0$, so that making EIG large enough cannot by itself drive the bias against to zero. The lesson is that controlling the bias against requires controlling the variance of the log relative belief ratio in addition to its mean, the EIG, and this is what the ratio criterion does.

Example 1 (EIG doesn't control bias against)**.** *Consider a random sample $w_1, \ldots, w_n \sim N(\theta, \sigma^2)$ where $\sigma^2 > 0$ is known and θ is unknown with a normal prior, $N(0, \tau^2)$. Write $\bar{w}$ for the sample mean, with distribution $\bar{w}|\theta \sim N(\theta, s^2)$, where $s^2 := \sigma^2/n$. Let ξ be a Bernoulli random variable with $Pr(\xi = 1) = q$, where q is known, and $B \sim N(0, \sigma_B^2)$ where $\sigma_B^2 > 0$, with ξ and B independent. Suppose we observe $y = \bar{w}$ if $\xi = 0$ and $y = B$ if $\xi = 1$. Hence we observe a normal mean with probability $1 - q$, but with probability q there is a failure, and in that case we observe a random variable that is uninformative about θ.*

Let us write $p_0(y|\theta) = \phi(y; \theta, s^2)$ for the density of y given $\theta, \xi = 0$, and $p_1(y) = \phi(y; 0, \sigma_B^2)$ for the density of y given $\xi = 1$. The likelihood is

$$p(y|\theta) = (1 - q)p_0(y \mid \theta) + qp_1(y),$$

and the prior predictive density of y, denoted $m(y)$, is

$$m(y) = (1 - q)m_0(y) + qp_1(y),$$

where $m_0(y) := \phi(y; 0, s^2 + \tau^2)$. Observe that the function $\log \frac{(a+c)}{(b+c)}$ is monotonic in c, taking the value $\log(a/b)$ at $c = 0$ and tending to zero as $c \to \infty$. We can write the log relative belief

as

$$Z = \log \frac{p(y|\theta)}{p(y)} = \log \frac{(1-q)p_0(y;\theta) + qp_1(y)}{(1-q)m_0(y) + qp_1(y)},$$

which has the form of $\log \frac{(a+c)}{(b+c)}$ *with* $a = (1-q)p_0(y;\theta)$, $b = (1-q)m_0(y)$ *and* $c = qp_1(y)$. *Hence* $0 \leq |Z| \leq |K|$ *where* $K = \log a/b = \log(p_0(y;\theta)/m_0(y))$, *and* Z *and* K *have the same sign. This last fact means we can write the bias against as*

$$\text{BA} = P(Z \leq 0) = P(K \leq 0).$$

Since K *is the log relative belief for a normal location problem with normal prior, we can compute it explicitly as*

$$K = \frac{1}{2}\log\left(1 + \tau^2/s^2\right) - \frac{(y-\theta)^2}{2s^2} + \frac{y^2}{2(s^2+\tau^2)}. \tag{24}$$

Next, write

$$\begin{aligned} P(K \leq 0) &= P(\xi = 0)P(K \leq 0|\xi = 0) + P(\xi = 1)P(K \leq 0|\xi = 1) \\ &= (1-q)P(K \leq 0|\xi = 0) + qP(K \leq 0|\xi = 1). \end{aligned} \tag{25}$$

We show in Appendix C that

$$P(K \leq 0|\xi = 1) \geq 1 - Cs(\log n)^{1/2} \tag{26}$$

where C *is a constant depending only on* τ *and* σ_B, *and*

$$P(K \leq 0|\xi = 0) \leq \sqrt{2}(s/\tau)^{1/2}. \tag{27}$$

Combining (25), (26) *and* (27), *and observing that* $s = \sigma/\sqrt{n} \to 0$ *as* $n \to \infty$, *we have the bias against satisfies*

$$P(Z \leq 0) = P(K \leq 0) \to q,$$

as $n \to \infty$. *So the bias against cannot be made to go to zero no matter how large* n *is.*

In Appendix C we also demonstrate that

$$\frac{1-q}{2}\log\left(1 + \frac{\tau^2}{s^2}\right) - H(q) \leq \text{EIG} \leq \frac{1-q}{2}\log\left(1 + \frac{\tau^2}{s^2}\right), \tag{28}$$

where $H(q) = -q\log q - (1-q)\log(1-q)$. *Recalling that* $s^2 = \sigma^2/n$, *we see that* $\text{EIG} = (1-q)/2\log n + O(1)$. *This establishes that* $\text{EIG} \to \infty$.

Finally let us deduce that both the EIG and its variance go to infinity with n, *which is why EIG does not control bias against in this example and why criteria such as* R *which consider*

this variance are more suitable. From (17)*, and since we have shown that* $\mathrm{BA} = q + o(1)$*, we have*

$$\frac{\mathrm{Var}(Z)}{\mathrm{EIG}^2} \geq \frac{q}{1-q} + o(1),$$

which gives

$$\mathrm{Var}(Z) \geq \left(\frac{q}{1-q} + o(1)\right) \mathrm{EIG}^2.$$

Hence $\mathrm{Var(Z)} \to \infty$ *with* n *since* $\mathrm{EIG} \to \infty$ *does, and* (17) *also allows us to deduce that for the ratio criterion* R*,* $\limsup R^2 \leq (1-q)/q$*, and hence from* (18)*,* $\liminf \widetilde{\mathrm{BA}} \geq q$*, as should be the case because* $\widetilde{\mathrm{BA}}$ *is an upper bound on* BA*.*

The example above is unusual in that the posterior fails to concentrate at the true value with probability q. However, if a Bernstein-von Mises theorem holds, then we can say something about the EIG, BA and ratio criteria asymptotically. As in the example above, consider for the moment a fixed design (suppressing x) with $y_1, \ldots, y_n$ independent and identically distributed from a distribution with parameter $\theta \in \mathbb{R}^d$ having prior density $p(\theta)$.

Assume the conditions required for Theorem 2.1 of Clarke and Barron (1990) to hold, together with equality of the two information matrices appearing in the theorem, and we write the information matrix as $\mathcal{I}(\theta)$. We also assume below it is valid to average limits over the prior. Then the log relative belief ratio satisfies

$$Z = \frac{d}{2}\log n + c(\theta) - \frac{W_n}{2} + o_p(1), \tag{29}$$

where $c(\theta) := \frac{1}{2}\log\det\mathcal{I}(\theta) - \frac{d}{2}\log(2\pi) - \log p(\theta)$ and $W_n := S_n^\top \mathcal{I}(\theta)^{-1} S_n$, with $S_n := n^{-1/2}\sum_{i=1}^n \nabla_\theta \log p(y_i \mid \theta)$ the scaled score. In this form, (29) is the second limit of Theorem 2.1 of Clarke and Barron (1990) rearranged, and the $o_p(1)$ term converges to zero in L^1 given θ. Conditionally on almost every θ, $W_n \to \chi^2_d$ in distribution by the central limit theorem for the score, and since the limit does not depend on θ, W_n is asymptotically independent of θ. Averaging over the prior gives the expansion

$$\mathrm{EIG} = \frac{d}{2}\log\frac{n}{2\pi e} + \frac{1}{2}E\{\log\det\mathcal{I}(\theta)\} - E\{\log p(\theta)\} + o(1) \to \infty,$$

an expansion of the same form as the one used by Long et al. (2013) for fast approximate computation of the EIG. In addition, $Z \to \infty$ in probability, so $\mathrm{BA} \to 0$. Under further regularity assumptions, $\mathrm{Var}(Z) \to \mathrm{Var}\{c(\theta)\} + d/2 < \infty$, so $R \sim (\mathrm{Var}\{c(\theta)\} + d/2)^{-1/2}\,\mathrm{EIG}$. This means that we cannot have $\mathrm{EIG} \to \infty$ unless $R \to \infty$ as well. The upper bound $\widetilde{\mathrm{BA}} = (1+R^2)^{-1} \to 0$, although at the slow rate of $(\log n)^{-2}$.

The discussion above does not directly apply to the policy-based designs used in the rest of the paper, where the design policy η_φ is for a fixed number of experiments T (Section 3.5), so that it isn't clear what n is in the asymptotic results. It is natural to treat the whole rollout of the T experiments given φ as a single observation, and to have n independent replicates of this. Write the independent trajectories $y^{(1)}, \ldots, y^{(n)}$ as $y^{(j)} = (y_1^{(j)}, \ldots, y_T^{(j)}) \sim p(y \mid \varphi, \theta)$ given a common θ. Since these are iid given θ, with $p(y \mid \varphi, \theta)$ playing the role of the density for a single observation, the iid result above applies directly with n now the number of replicate experiments. Writing $\mathcal{I}(\varphi, \theta)$ for the information matrix of one replicate trajectory under φ, and assuming the conditions of Theorem 2.1 of Clarke and Barron (1990) hold together with equality of its two information matrices, we obtain the expansion (29) with θ, $\mathcal{I}(\theta)$ and S_n reinterpreted as above and $S_n := n^{-1/2} \sum_{j=1}^n \nabla_\theta \log p(y^{(j)} \mid \varphi, \theta)$. Here $c(\theta)$ becomes $c(\varphi, \theta)$, defined in the same way with $\mathcal{I}(\varphi, \theta)$ in place of $\mathcal{I}(\theta)$. Consequently $\mathrm{EIG}(\varphi) \to \infty$, $\mathrm{BA}(\varphi) \to 0$ and $R(\varphi) \to \infty$ as $n \to \infty$ for fixed φ, with $R(\varphi) = \{\frac{d}{2} \log n + a(\varphi)\}/s(\varphi) + o(1)$, where $a(\varphi)$ is the policy-dependent constant in the EIG expansion and $s(\varphi)^2 := \mathrm{Var}\{c(\varphi, \theta)\} + d/2$. Thus for large n the EIG policy is governed by $a(\varphi)$ alone, while the BA/ratio policy is governed by $s(\varphi)$, favouring policies for which $\log\det \mathcal{I}(\varphi, \theta)$ is less variable over the prior. This asymptotic result is used here only to illustrate why the ratio criterion is sensitive to the variability of $\log\det \mathcal{I}(\varphi, \theta)$ over the prior, unlike the EIG criterion. Training our policies uses $n = 1$, and in this case we need to study $\mathrm{Var}(Z(y, \varphi, \theta))$ directly. Another asymptotic regime that could be studied would let $T \to \infty$, but this is complicated because of the dependence induced by the adaptive design.

3.5 Policy-based design with bias against

Throughout the rest of the paper we work with the policy formulation for amortized sequential design, where for a sequence of experiments indexed by $t = 1, \ldots, T$, we make a design choice x_t at stage t which is determined from $\mathcal{I}_{t-1}$ by a parametrized policy η_φ. As in the EIG discussion above we write $p(y|\varphi, \theta) = \prod_{t=1}^T p(y_t|x_t = \eta_\varphi(\mathcal{I}_{t-1}), \theta)$ and $p(y|\varphi) = E_{\theta \sim p(\theta)}(p(y|\varphi, \theta))$, and we let $Z(y, \varphi, \theta) := \log p(y|\varphi, \theta) - \log p(y|\varphi)$. With this notation the optimization target we consider for optimizing the upper bound $\widetilde{\mathrm{BA}}(x)$ is

$$L_\alpha(\varphi) = E(Z(y, \varphi, \theta)) - \alpha \, \mathrm{Var}(Z(y, \varphi, \theta)), \tag{30}$$

and the ratio criterion is

$$R(\varphi) = \frac{E(Z(y, \varphi, \theta))}{\sqrt{\mathrm{Var}(Z(y, \varphi, \theta))}}. \tag{31}$$

The bound (21), the ratio criteria (19) and (22), and the correspondence between ratio and penalized criteria all carry over to the policy formulation with φ in place of x, since these results depend on the design only through the mean and variance of the log relative belief ratio. See Remark 1 in Appendix B.

4 Computational issues

Both $L_\alpha(\varphi)$ and $R(\varphi)$ contain the intractable prior predictive density $p(y|\varphi)$ and hence must be approximated. We first describe a Monte Carlo estimator of the objective, of the kind employed by Foster et al. (2021) for EIG estimation, and then describe how it is used within a stochastic gradient optimization in which the Monte Carlo samples are refreshed at every step.

4.1 Monte Carlo estimator of the objective

Assume that a simulation from the joint Bayesian model can be expressed as a deterministic differentiable transformation of random variables (η, ρ), written as $(\theta, y) = (G_1(\eta), G_2(\eta, \rho, \varphi))$. It is assumed that the distribution of (η, ρ) does not depend on φ. For example, they could be standard normal variates. This kind of reparametrization is available for most of the priors and likelihoods used in our examples.

Draw a set of J independent (ρ, η) pairs, $(\eta^{(j)}, \rho^{(j)})$, $j = 1, \ldots, J$, and a further set of L independent draws for η, $\widetilde{\eta}^{(l)}$, $l = 1, \ldots, L$. Define $\theta^{(j)} = G_1(\eta^{(j)})$, $y^{(j)} = G_2(\eta^{(j)}, \rho^{(j)}, \varphi)$, and $\widetilde{\theta}^{(l)} = G_1(\widetilde{\eta}^{(l)})$. We estimate $p(y^{(j)}|\varphi)$ by

$$\widehat{p}(y^{(j)}|\varphi) := \frac{1}{L+1}\left\{p(y^{(j)}|\varphi, \theta^{(j)}) + \sum_{l=1}^{L} p(y^{(j)}|\varphi, \widetilde{\theta}^{(l)})\right\}, \tag{32}$$

and set

$$\widehat{Z}^{(j)}(\varphi) := \log p(y^{(j)}|\varphi, \theta^{(j)}) - \log \widehat{p}(y^{(j)}|\varphi).$$

Writing

$$\overline{Z}(\varphi) := \frac{1}{J}\sum_{j=1}^{J} \widehat{Z}^{(j)}(\varphi), \qquad S(\varphi) := \frac{1}{J-1}\sum_{j=1}^{J}\left(\widehat{Z}^{(j)}(\varphi) - \overline{Z}(\varphi)\right)^2,$$

our Monte Carlo objective for fixed $\alpha \geq 0$ is

$$\widehat{L}_\alpha(\varphi) = \overline{Z}(\varphi) - \alpha\, S(\varphi). \tag{33}$$

with the corresponding estimate of the ratio criterion being $\widehat{R}(\varphi) := \overline{Z}(\varphi)/\sqrt{S(\varphi)}$. The reparametrization makes $\widehat{L}_\alpha(\varphi)$ and $\widehat{R}(\varphi)$ differentiable in φ for a fixed realization of the noise variables $\eta^{(j)}, \rho^{(j)}, \widetilde{\eta}^{(l)}$, so that gradients can be obtained by automatic differentiation and the policy can be efficiently trained by stochastic gradient ascent. The mean term $\overline{Z}(\varphi)$ is a direct Monte Carlo estimate of the lower-bound estimator (10), so that with $\alpha = 0$ (33) reduces to an estimate of the PCE bound similar to the one used in Foster et al. (2021). For $\alpha > 0$, the additional term penalizes the Monte Carlo variance of the estimated log relative belief ratio.

4.2 Stochastic optimization with refreshed Monte Carlo samples

At each stochastic-gradient update we draw a fresh Monte Carlo batch, form the estimator (33), take one gradient step, and then discard the batch. Thus the optimization uses refreshed Monte Carlo approximations of the population objective, rather than a single fixed randomized approximation.

More precisely, with current iterate $\varphi^{(s)}$ and step size γ_s, we update

$$\varphi^{(s+1)} = \varphi^{(s)} + \gamma_s \nabla_\varphi \widehat{L}_\alpha(\varphi)\big|_{\varphi=\varphi^{(s)}}, \tag{34}$$

where the Monte Carlo variables entering $\widehat{L}_\alpha$ are drawn independently at each step s. In our implementation the outer batch of size J can be accumulated over several smaller minibatches, which reduces memory requirements. The gradients of the minibatch contributions are accumulated before the parameter update. When more than one minibatch is used, the variance term $S(\varphi)$ is computed within each minibatch and averaged across minibatches. We use the unbiased form of the sample variance estimator in all cases, so this does not change the quantity being estimated, which will be the variance of the log relative belief ratio when $L \to \infty$. Using a single minibatch of size J recovers the expression given above.

The gradient of $\widehat{L}_\alpha(\varphi)$ is not an unbiased estimator of $\nabla_\varphi L_\alpha(\varphi)$ for finite J and L. The optimization (34) is therefore a stochastic gradient scheme with biased gradients, with the bias vanishing as $J, L \to \infty$. Despite this bias, we find that small values of J and L are sufficient to learn good policies, while keeping computational and memory requirements low. A similar approximation, in which a biased but inexpensive gradient is used based on a batch approximation in place of a more costly unbiased one, was used successfully by Chen et al. (2025) in a variational inference context. We use the Adam optimizer (Kingma and Ba, 2014) for the updates (34). In some examples we also use periodic model checkpointing for numerical stability. The example-specific training details are given in Appendix E, where we also explore sensitivity of the optimization to the tuning parameters in the stochastic gradient algorithm.

4.3 Choice of the penalty parameter

The discussion in Section 3.2 shows that the penalty parameter α has an evidential interpretation. Varying α traces out designs with different degrees of conservativeness in controlling misleading evidence. Equivalently, different values of α can be viewed as corresponding to different threshold events of the form $\{Z \leq t\}$ whose bounds are being tightened. In the examples below we consider a grid of possible values for α,

$$\alpha \in \{0.5, 0.75, 1.0, 1.25, 1.5\}.$$

For each value of α, we train a policy by maximizing the Monte Carlo objective (33). The resulting policies are not interpreted as ordinary hyperparameter perturbations of a single objective. Rather, they form a path of designs from less conservative to more conservative policies, with larger α placing greater emphasis on reducing the variability of the log relative belief ratio. Finally, if we wish to choose a single α, we do so by selecting the α attaining the highest estimated ratio $\widehat{R}(\varphi) := \overline{Z}(\varphi)/\sqrt{S(\varphi)}$ on average, evaluated on a random batch sample. This corresponds to the tightest bound on the bias against.

For the source-location and hyperbolic-discounting examples of Sections 5.1 and 5.2 respectively, we use the policy-network architecture of Foster et al. (2021). The network embeds each previous design-outcome pair and aggregates these embeddings, so that the proposed design depends on the accumulated experimental history rather than on an arbitrary ordering of the history elements. For the pizza choice example of Section 5.3, we use the same history-aggregation structure as in the hyperbolic-discounting example, with input and output dimensions and hyperparameters adjusted to the design space.

4.4 Gumbel-Softmax relaxation

Our discussion so far assumes that the outcome y can be written as a differentiable function $y = G_2(\eta, \rho, \varphi)$ with η, ρ random variables having distributions not depending on φ, so that $\widehat{L}_\alpha(\varphi)$ depends smoothly on φ for fixed draws of (η, ρ). When y is discrete this does not hold (as in the hyperbolic discounting example of Section 5.2 below, where the response is Bernoulli). For these examples we use differentiable relaxations during training (Jang et al., 2016; Maddison et al., 2016). Bernoulli or categorical observations are replaced by Gumbel-Softmax variables, so that the simulated outcome is a differentiable function of the policy parameter and random noise. The corresponding approximation is used in the Monte Carlo objective during training. A description of the Gumbel-softmax trick is given in Appendix D.

The Gumbel-softmax relaxation is used only for optimization and policy net training. For the final design-trace evaluations, we use the original discrete simulator. Thus the learned pol-

icy is trained using a smooth approximation that enables gradient-based optimization, while the reported adaptive design traces are generated under the intended discrete data-generating model. The precise simulator, likelihood and any example-specific training stabilization are described in the individual examples.

5 Examples

We consider three examples. The first two are based on examples in Foster et al. (2021), while the third is a discrete choice experiment on a pizza choice. In each example we compare a policy trained for EIG with policies trained using the BA objective (33). For the location-finding example the BA policies are trained from random initialization. For the hyperbolic-discounting and pizza choice examples, we use a common EIG warm start. Starting from this same warm start, we then continue fine-tuning for additional epochs under either the EIG objective or the BA objective, resulting in the EIG and BA policies respectively that are compared in the Figures.

For visualization of adaptive design traces, we use the same general procedure in all three examples. For a fixed parameter value θ, each policy is run repeatedly, producing a sample of stochastic design traces. We treat each trace as an ordered sequence of designs. If $X^{(m)} = (x_1^{(m)}, \ldots, x_T^{(m)})$ denotes the mth trace, we stack it into a vector $v^{(m)} = \mathrm{vec}(X^{(m)})$ and define the representative trace by the geometric median

$$\widehat{v} = \arg\min_{v} \sum_{m} \|v - v^{(m)}\|_2.$$

The vector $\widehat{v}$ is then reshaped to obtain the representative design trace. We compute this geometric median using the Weiszfeld iteration algorithm (Weiszfeld, 1937). The use of representative traces over many simulations for the same θ is used to remove noise in order to make the differences between the design policies easier to recognize and understand.

To compare policies trained by different objectives, we compute the dynamic time warping (DTW) distance between their representative traces (Giorgino, 2009). We then sample a set of parameter values from the prior and display representative traces for parameter values corresponding to selected $(0\%, 20\%, 40\%, 60\%, 80\%, 100\%)$ quantiles of the DTW distance, together with several manually chosen parameter values that illustrate different inferential regimes. The detailed hyperparameters for training and visualization are presented in Appendix E. Hereafter we call the policy trained by optimizing objective (10) the EIG policy, and the policy trained by optimizing objective (33) the BA policy. For the pizza choice example in Section 5.3, the design at each step contains two-dimensional attributes of three pizza options. We compute

the representative trace separately for each option here. We sum the three DTW distances for the options to get the overall DTW distance for selecting parameter values based on the quantiles. All codes and results are saved in `https://github.com/DZCQs/DAD-bias_against`.

5.1 Source location finding

We consider first the source location finding example given in Foster et al. (2021), based on a related model in Sheng and Hu (2005). There are two sources in two-dimensional space, which send signals where the intensity decays with distance according to an inverse squared law. The locations of the first and second source will be denoted by $\Theta_1 = (\theta_1, \theta_2)^\top$ and $\Theta_2 = (\theta_3, \theta_4)^\top$, with the unknown parameter to be inferred $\theta = (\Theta_1^\top, \Theta_2^\top)^\top$. A multivariate standard normal prior is assumed for θ. We can take T measurements at different times from a sensor that can be moved at each time. The sensor location is a design variable, so we have to choose the location $x_t = (x_{t1}, x_{t2})^\top$ adaptively at each time given the previous locations and measurements. At time t we observe y_t, a noisy measurement informative about the total log-intensity of the signal from the two sources. Following Foster et al. (2021), the total intensity at sensor location x is

$$\mu(\theta, x) = 0.1 + \sum_{k=1}^{2} \frac{1}{10^{-4} + \|\Theta_k - x\|_2^2},$$

and we model the measurement on the log-intensity scale with additive heavy-tailed noise,

$$y_t = \log \mu(\theta, x_t) + \sigma\, \varepsilon_t, \qquad \varepsilon_t \sim t_\nu,$$

where ε_t has a Student-t distribution with $\nu = 3$ degrees of freedom and $\sigma = 0.7$. The heavy-tailed observation noise differs from the Gaussian noise used by Foster et al. (2021) and makes the inference problem more susceptible to occasional misleading observations, which is the regime of interest for the bias against criterion.

Figure 1 compares the EIG policy with BA policies. For each trained policy, we run a Monte Carlo evaluation batch and compute the sequential estimates $\overline{Z}(\varphi)$, $S(\varphi)$, and $\widehat{R}(\varphi) = \overline{Z}(\varphi)/\sqrt{S(\varphi)}$ after each experimental step. The EIG policy gives the largest value of $\overline{Z}(\varphi)$, as expected, since it is trained directly to maximize the EIG lower bound. The BA policies sacrifice some value of $\overline{Z}(\varphi)$, but reduce the estimated variability $S(\varphi)$ of the log relative belief ratio. As a result, for suitable values of α the BA policy achieves a larger ratio $\widehat{R}(\varphi)$ over most experimental steps. This is consistent with the comparison in Section 3.4: in this example the BA objective changes the policy mainly through the design-induced variability of Z, reducing $S(\varphi)$ while accepting some reduction in $\overline{Z}(\varphi)$.

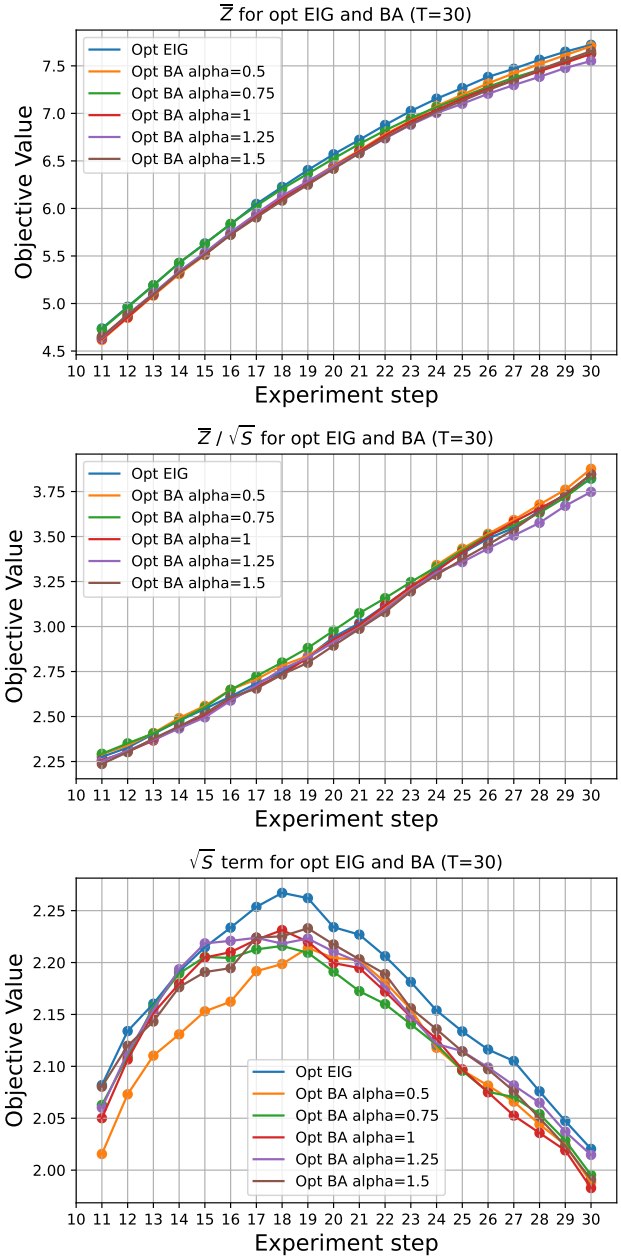


Figure 1: Evaluated $\overline{Z}$ (top), $\overline{Z}/\sqrt{S}$ (middle) and $\sqrt{S}$ (bottom) over $T = 30$ experimental steps for location finding example for EIG policy and BA policies with $\alpha = 0.5, 0.75, 1.0, 1.25, 1.5$.

Figure 2 plots representative traces for different parameter values chosen from simulations from the prior based on quantiles of DTW distance as discussed above. For the BA policy we use $\alpha = 0.75$, selected according to the procedure described in Section 4.3. The selection of the parameter values by quantiles of DTW distance ensures that the cases range from those where the policies behave most similarly, to those where they are most different. Both policies adapt to the underlying source locations and tend to take smaller steps when the sensor is close to the signal sources. Although both policies show similar behaviour, in 5 of the 6 plots the BA policy designs explore the design space more widely. Figure 3 compares the policies for two manually chosen parameter configurations, one where the two sources are far apart, and one where they are close together. When the sources are close together, the EIG policy is more dispersed, whereas for the BA policy the opposite behaviour is observed.

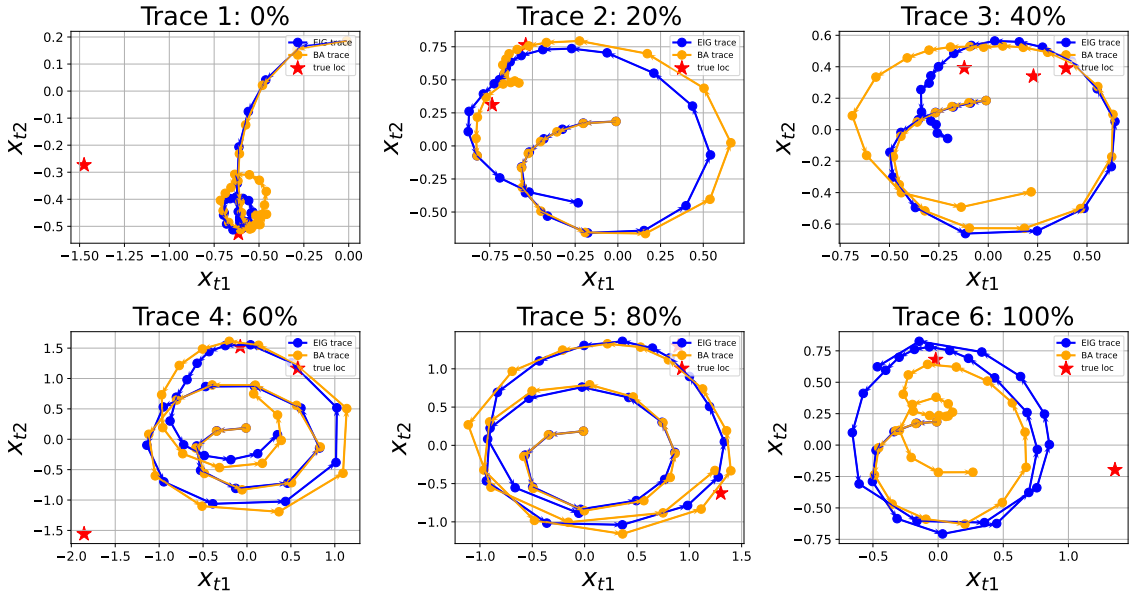


Figure 2: Representative design traces for the source-location example. For each sampled value of θ, we run the EIG and BA policies repeatedly and summarize the resulting stochastic traces by geometric medians. The panels show parameter values for which the DTW distance between the representative traces corresponds to the selected quantiles $(0\%, 20\%, 40\%, 60\%, 80\%, 100\%)$ over a sample from the prior. The red star-like dots are the true signal sources. The blue arrows are representative design traces from the EIG policy, and the orange arrows are representative design traces from the BA policy.

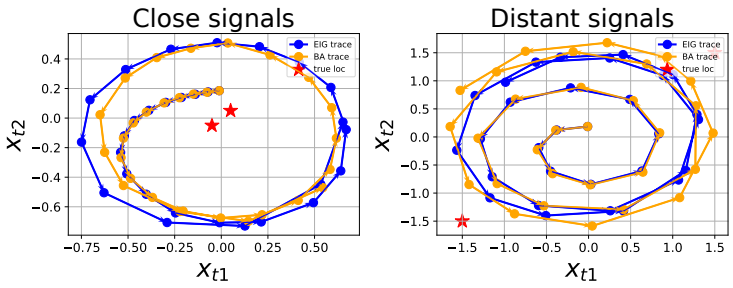


Figure 3: Representative design traces for the source-location example at manually selected parameter values. For each value of θ, we run the EIG and BA policies repeatedly and summarize the resulting stochastic traces by geometric medians. The red star-like dots are the true signal sources. The blue arrows are representative design traces from the EIG policy, and the orange arrows are representative design traces from the BA policy.

5.2 Hyperbolic temporal discounting

Next we consider a design problem for a hyperbolic discounting model discussed in Foster et al. (2021). The purpose of experiments like this one is to study how much people value immediate compared to delayed rewards. The example illustrates use of our approach in a setting where the data are discrete. Participants in a study were asked to choose between the two propositions "£R today" or "£100 in D days", with values V_0 and V_1 respectively, where $V_0 = R$ and $V_1 = 100/(1 + kD)$ and $k > 0$ is a discounting parameter. The design variables are $x = (R, D)$, the immediate reward and the delay. We observe $y \in \{0, 1\}$, equal to 1 if the respondent chooses the immediate reward V_0 rather than the delayed reward V_1, and it is assumed that

$$P(y = 1 \mid \theta, x) = \epsilon + (0.5 - \epsilon) \left\{ 1 + \operatorname{erf}\left(\frac{V_0 - V_1}{a} \right) \right\},$$

where $\theta = (k, a)$ is the model parameter, erf is the error function, $a > 0$ controls the sharpness of the choice probability, and the lapse rate is fixed at $\epsilon = 0.01$, so that the choice probability ranges between 0.01 and 0.99. We place a log-normal prior on the discount parameter, $\log k \sim N(-4.25, 1.5^2)$, and a half-normal prior on the sharpness, $a \sim N_+(0, 2^2)$, with k and a independent a priori.

In the policy-based formulation, the network does not output the design variables $x = (R, D)$ directly. Instead it outputs two unconstrained real values which are transformed onto the valid ranges of the reward and delay. Writing (r, d) for the raw network outputs, the immediate reward is obtained as $R = 100\, F_3(r)$, where F_3 is the cumulative distribution function of a Student-t distribution with three degrees of freedom, mapping r into the interval

$(0, 100)$. The delay is obtained as $D = \exp\{d - \mathrm{E}(\log k)\}$, where $E(\log k) = -4.25$, which centres the delay on a scale commensurate with the discount rate. We consider, similar to Foster et al. (2021), an adaptive design for this problem with a sequence of $T = 20$ steps, choosing the design variables and making an observation y_t at each step, $t = 1, \ldots, T$.

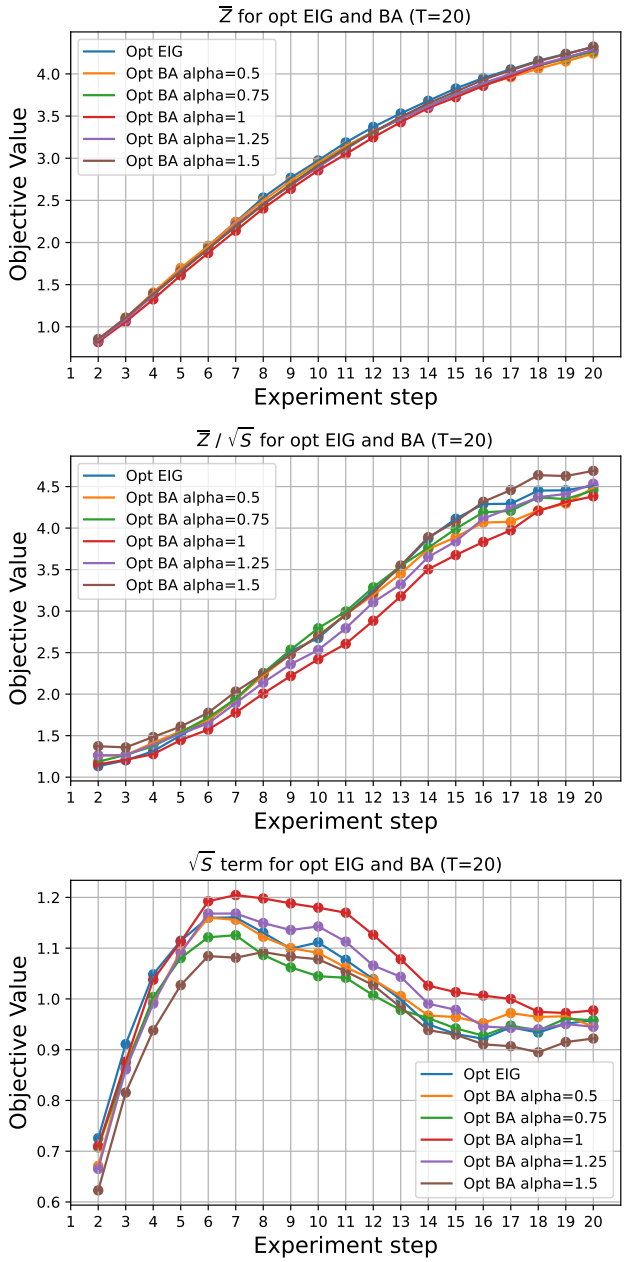


Figure 4: Evaluated $\overline{Z}$ (top), $\overline{Z}/\sqrt{S}$ (middle) and $\sqrt{S}$ (bottom) over $T = 20$ experimental steps for hyperbolic discounting example for EIG policy and BA policies with $\alpha = 0.5, 0.75, 1.0, 1.25, 1.5$.

Figure 4 compares the EIG policy with BA policies trained at different values of α. Similar

to the location finding example, the top panel shows the Monte Carlo estimate $\overline{Z}(\varphi)$, the middle panel shows the ratio $\widehat{R}(\varphi) = \overline{Z}(\varphi)/\sqrt{S(\varphi)}$ and the bottom panel shows the estimated standard deviation $\sqrt{S(\varphi)}$, all evaluated sequentially over the $T = 20$ experimental steps. The values of $\overline{Z}(\varphi)$ increase steadily for all policies and are very similar across the EIG and BA policies. The BA objective does not lead to a substantial loss in the EIG lower-bound estimate. The BA policies with $\alpha = 0.5, 0.75$ and 1.5 have smaller values of $\sqrt{S(\varphi)}$ over most experimental steps, and this reduction in variability leads to larger values of $\widehat{R}(\varphi)$, especially in the later stages of the experiment. For the BA policies, the expected log relative belief ratio is similar to that of the EIG policy, but its variability is reduced.

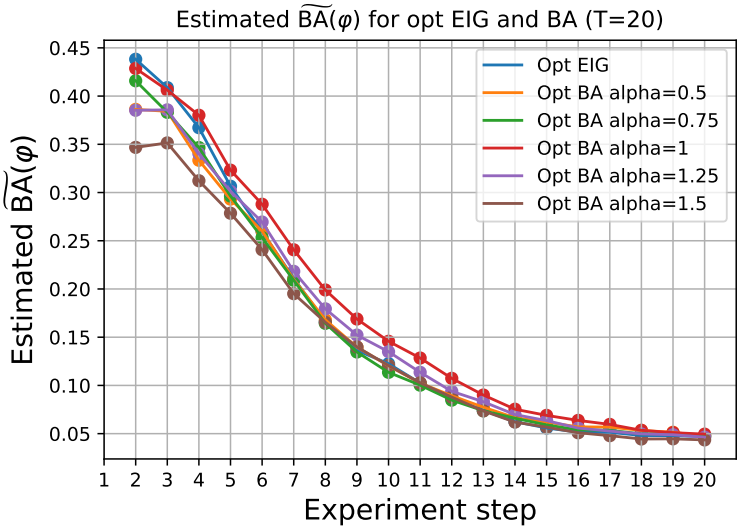


Figure 5: Estimated upper bound on bias against for the hyperbolic-discounting example. The plotted quantity is Estimated $\widetilde{BA}(\varphi) = \{1 + \widehat{R}(\varphi)^2\}^{-1}$, where $\widehat{R}(\varphi) = \overline{Z}(\varphi)/\sqrt{S(\varphi)}$ is evaluated sequentially over the $T = 20$ experimental steps. Smaller values correspond to a tighter upper bound on the bias against.

The corresponding effect on the estimated bias-against upper bound is shown directly in Figure 5, which presents the same comparison on the scale of the bias-against upper bound. Since the bound is a decreasing transformation of $\widehat{R}(\varphi)$, policies with larger values of the ratio criterion in Figure 4 correspond to smaller values in Figure 5. The BA policies reduce the estimated upper bound on bias against compared to the EIG policy over the $T = 20$ experimental steps.

Figures 6 and 7 show the same comparison on the subjective value scale. For a proposed design $x_t = (D_t, R_t)$ and parameter value $\theta = (k, a)$, the plotted quantity, the perceived value

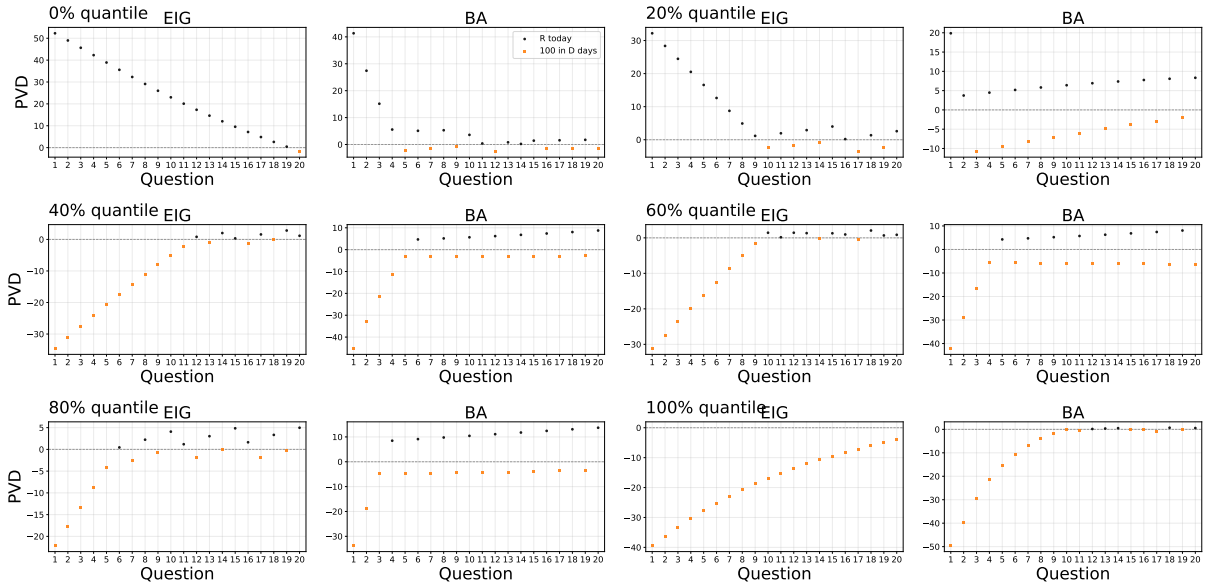


Figure 6: Representative subjective-value traces for the hyperbolic discounting example. The panels show parameter values for which the DTW distance between the representative traces corresponds to the selected quantiles $(0\%, 20\%, 40\%, 60\%, 80\%, 100\%)$ over a sample from the prior. Black dots indicate tasks for which "R pounds today" has higher subjective value, while orange dots indicate tasks for which "100 pounds in D days" has higher subjective value. The y-axis denotes the perceived value difference (PVD).

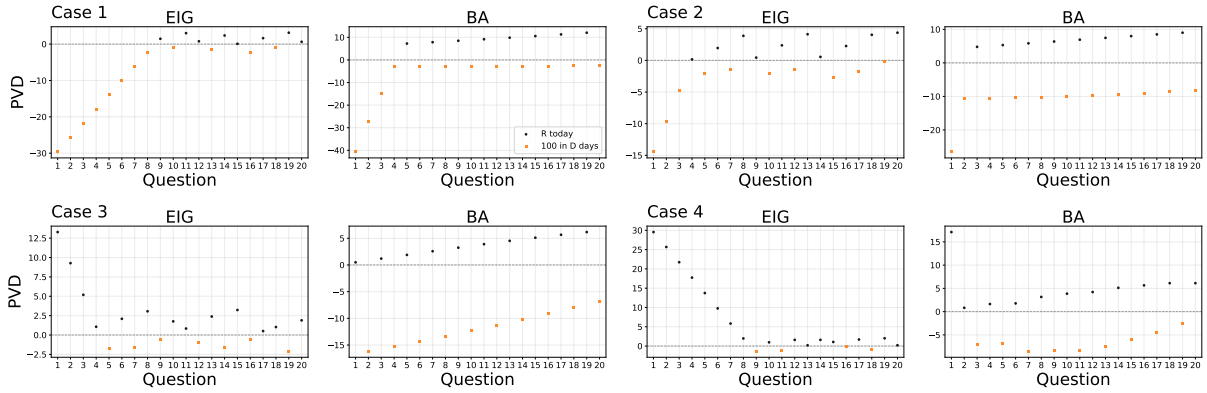


Figure 7: Representative subjective-value traces for the hyperbolic discounting example at manually selected parameter values. Black dots indicate tasks for which "R pounds today" has higher subjective value, while orange dots indicate tasks for which "100 pounds in D days" has higher subjective value. The y-axis denotes the perceived value difference (PVD). Parameter values (k, a) for Case 1/2/3/4 (Top left/Top right/Bottom left/Bottom right): $(0.005, 0.5)/(0.015, 1.0)/(0.05, 2.0)/(0.10, 3.0)$.

difference (PVD), is

$$R_t - \frac{100}{1 + kD_t}.$$

If this is positive, it means that the immediate reward has the larger subjective value, whereas if it is negative it means that the delayed reward has the larger subjective value. Values close to zero correspond to difficult questions, for which the choice probability is close to one half. In the plots, the BA policy generally shows larger changes in the perceived value difference at consecutive experimental steps, often alternating between design choices with positive and negative subjective value.

5.3 A discrete choice experiment

Finally, we consider an adaptive design problem for a stated-preference discrete choice experiment to infer an individual's optimal option profile and context-dependent choice behavior. The motivating setting here consists of a series of choice tasks in which participants select their most preferred option from three pizzas varying in spiciness and cheesiness. Within a choice task, pizza j's profile is

$$x_j = (x_{j1}, x_{j2})^\top \in [0,1]^2, \qquad j \in \{1,2,3\}.$$

where the two coordinates denote the degree of spiciness and cheesiness, respectively. The full design is

$$x = (x_1^\top, x_2^\top, x_3^\top)^\top \in [0,1]^6.$$

Preference for a specific flavor is unlikely to be monotonic. Instead, it may peak at a moderate intensity and decline at either extreme. Therefore, we apply an ideal-point model (Cooper and Nakanishi, 1983; Xu and Davenport, 2020), which assumes that an option's subjective value (utility, hereafter) increases if it approaches the ideal profile in attribute space. In addition, we relax the above assumption on a fully rational decision maker by incorporating context-dependent effects into the utility specification (Rooderkerk et al., 2011). The most ubiquitous context effect is the attraction effect whereby an option becomes more attractive (i.e., higher utility) when a similar but inferior alternative exists. To set up, the unknown parameter vector for our discrete choice model is

$$\theta = (\mu_1, \mu_2, \beta, \gamma, \lambda),$$

where $\mu = (\mu_1, \mu_2)^\top$ is a participant's ideal point (i.e., most preferred pizza profile), $\beta > 0$ controls sensitivity to distance from the ideal point, γ controls the strength of the attraction effect, and $\lambda > 0$ controls the locality of the attraction effect. Therefore, pizza j's utility

consists of a baseline component determined by its squared Euclidean distance from the ideal point μ, adjusted by the attraction effect $A_j(x,\theta)$ induced by the other options, which is

$$V_j(x,\theta) = -\beta\|x_j - \mu\|_2^2 + \gamma A_j(x,\theta),$$

and the choice probabilities are given by the standard softmax function,

$$P(y = j \mid x,\theta) = \frac{\exp\{V_j(x,\theta)\}}{\sum_{\ell=1}^{3}\exp\{V_\ell(x,\theta)\}}, \qquad j \in \{1,2,3\}.$$

Specifically, $A_j(x,\theta)$ captures the total attraction effect of the remaining options k on option j, with each option's influence decaying exponentially at a rate of λ with its distance from j.

$$A_j(x,\theta) = \sum_{k\neq j}\exp\{-\lambda\|x_j - x_k\|_2^2\}D_{jk}^{\text{hard}}(x,\theta),$$

where $D_{jk}^{\text{hard}}(x,\theta)$ is the indicator that pizza k triggers the attraction effect when evaluating the pizza j's utility. Only when option k is dominated by the target option j on each attribute (i.e., further from the optimal value μ_i) does its presence increase the attractiveness of option j. The hard dominance indicator is therefore

$$D_{jk}^{\text{hard}}(x,\theta) = \prod_{a=1}^{2}\mathbf{1}\{(x_{ka}-\mu_a)^2 - (x_{ja}-\mu_a)^2 > 0\}.$$

In total, we use $T = 12$ choice tasks for the experiment. The prior ranges are

$$\mu_1,\mu_2 \sim U(0,1), \qquad \log\beta \sim U(\log 5,\log 30), \qquad \gamma \sim U(-1,1), \qquad \log\lambda \sim U(\log 5,\log 80).$$

Training follows the common warm-start protocol described in Appendix E. For policy net training here, we use a smooth policy-output transformation to avoid gradient loss from clipping the design to the closed box $[0,1]^6$; meanwhile, a sigmoid function (with a fixed scale parameter of $c = 30$) is used as a smooth relaxation of the domination indicator $D_{jk}^{hard}(x;\theta)$ above. Further details are given in Appendix D.

Figure 8 compares the EIG policy with BA policies for the pizza choice example over $T = 12$ choice tasks. The top panel shows that all policies achieve very similar values of $\overline{Z}(\varphi)$. The EIG policy is slightly higher in the later stages, but the gap is small, so that training by the BA objective (33) does not substantially reduce the EIG lower-bound estimate. The main separation appears in the lower two panels. The BA policies, especially for $\alpha = 1.5$, reduce $\sqrt{S(\varphi)}$ over most experimental steps relative to the EIG policy. This lower variability leads to larger values of the ratio $\widehat{R}(\varphi) = \overline{Z}(\varphi)/\sqrt{S(\varphi)}$, particularly after the first few choice tasks. Thus, in the pizza example, optimizing the BA objective gives a similar mean log relative

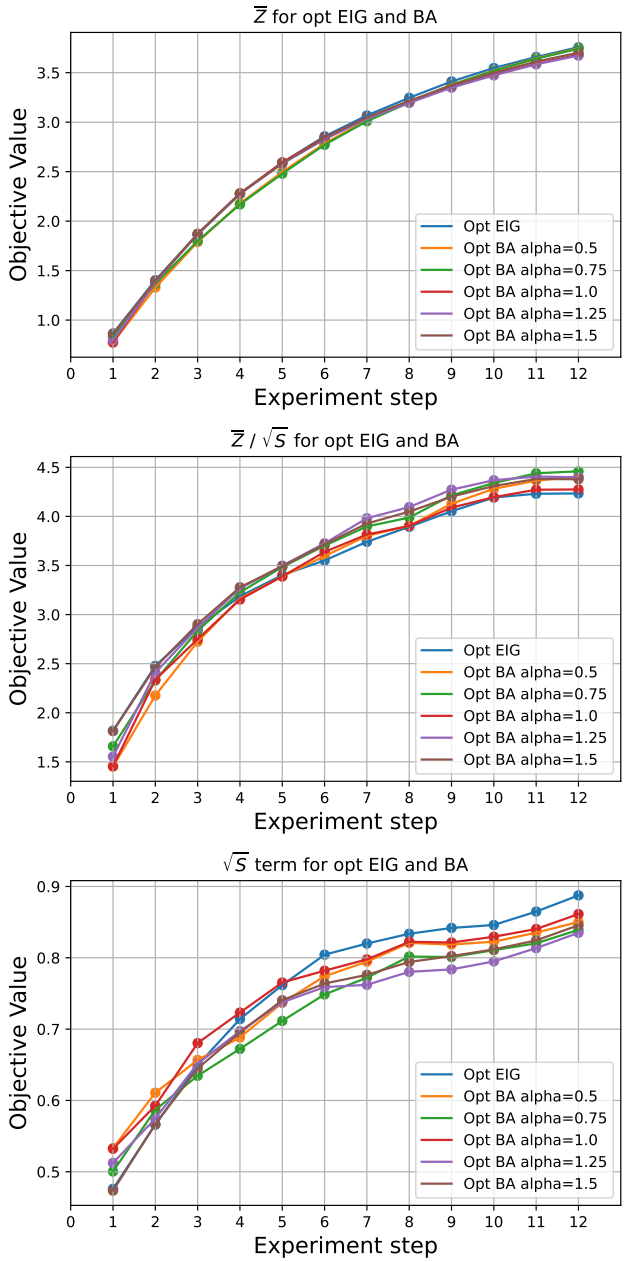


Figure 8: Evaluated $\overline{Z}$ (top), $\overline{Z}/\sqrt{S}$ (middle) and $\sqrt{S}$ (bottom) over $T = 12$ experimental steps for pizza example for EIG policy and BA policies with $\alpha = 0.5, 0.75, 1.0, 1.25, 1.5$.

belief but smaller variability, again illustrating the design-dependent variance effect discussed in Section 3.4. In terms of the ratio criterion, we choose $\alpha = 1.5$ as the BA policy for the design-trace evaluation. The difference between the learned policies is stable across alternative combinations of Monte Carlo evaluation sample sizes; see Figure 12 in Appendix E.

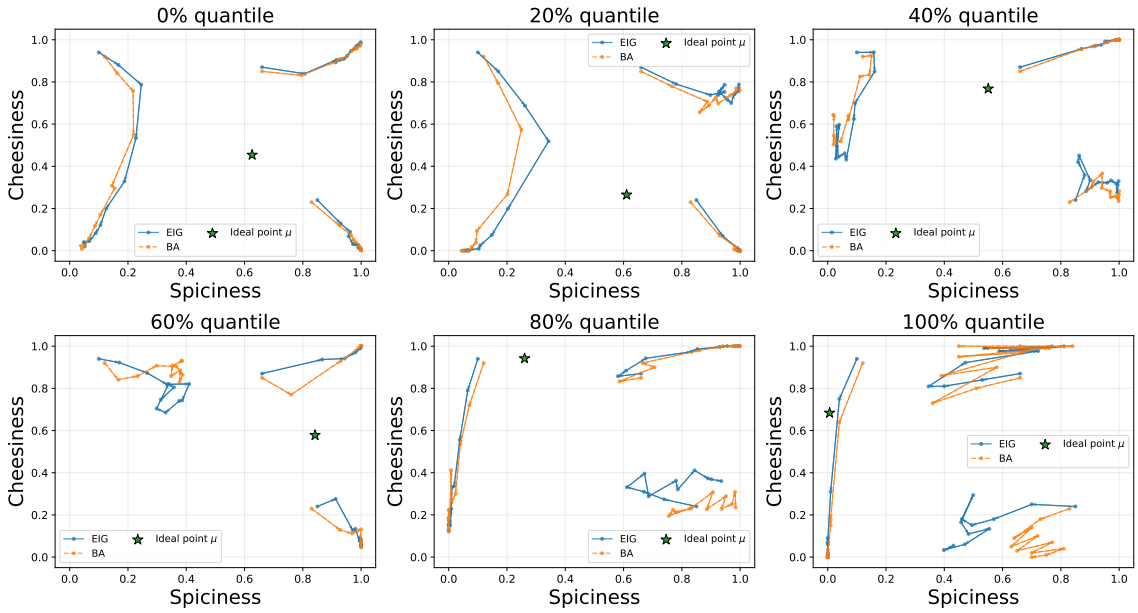


Figure 9: Representative design traces for the pizza choice example. For each sampled value of θ, we run the EIG and BA policies repeatedly and summarize the resulting stochastic traces by geometric medians. The panels show parameter values for which DTW distance (summed over options) between the representative traces corresponds to the selected quantiles $(0\%, 20\%, 40\%, 60\%, 80\%, 100\%)$ over a sample from the prior. The blue arrow curves are representative traces from the EIG policy, and the orange arrow curves are representative traces from the BA policy. Each panel contains three curves, corresponding to the three pizza options.

Figures 9 and 10 compare the representative pizza-design traces produced by the EIG and BA policies. Each point in a trace is one pizza option design in the spiciness-cheesiness space, and the three separated traces correspond to the adaptive design movement of the three pizza options across the $T = 12$ choice tasks. The ideal point μ is shown in each panel, so the traces can be interpreted relative to the participant type being simulated. The BA policy tends to move pizza options attributes slightly differently from those of EIG policy. Usually one of the three pizza options would have its BA policy design closer to the corner. The BA policy changes the design behavior by emphasizing boundary choice sets, while Figure 8 shows that this is accompanied by a smaller $\sqrt{S(\varphi)}$ and a similar value of $\overline{Z}(\varphi)$.

Figure 11 and Table 1 empirically illustrate the differences between the EIG and BA

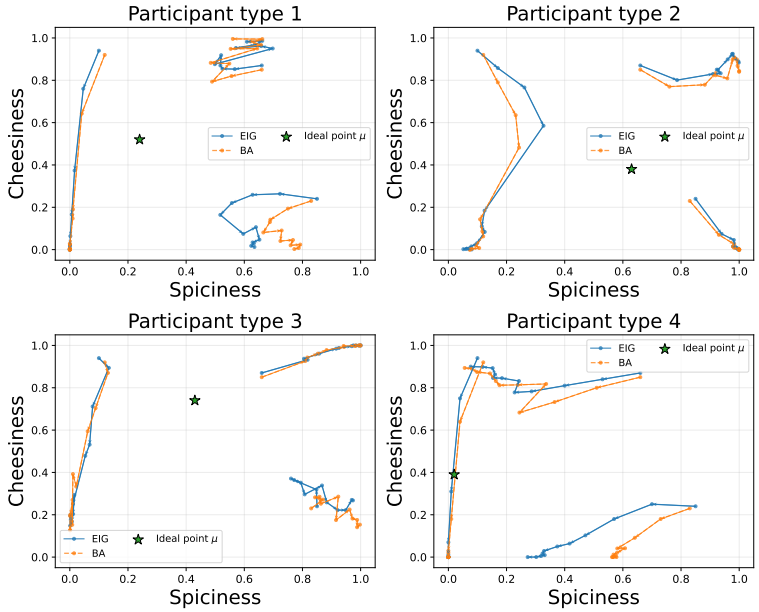


Figure 10: Representative design traces for the pizza choice example at manually selected parameter values. For each value of θ, we run the EIG and BA policies repeatedly and summarize the resulting stochastic traces by geometric medians for each option separately. The blue arrow curves are representative traces from the EIG policy, and the orange arrow curves are representative traces from the BA policy. Each panel contains three curves, corresponding to the three pizza options. Parameter values (β, γ, λ) for Pizza customer participant type 1/2/3/4 (Top left/Top right/Bottom left/Bottom right): $(14.50, 0.43, 57.00)/(16.00, 0.82, 32.00)/(9.50, 0.14, 6.00)/(17.00, 0.74, 78.00)$.

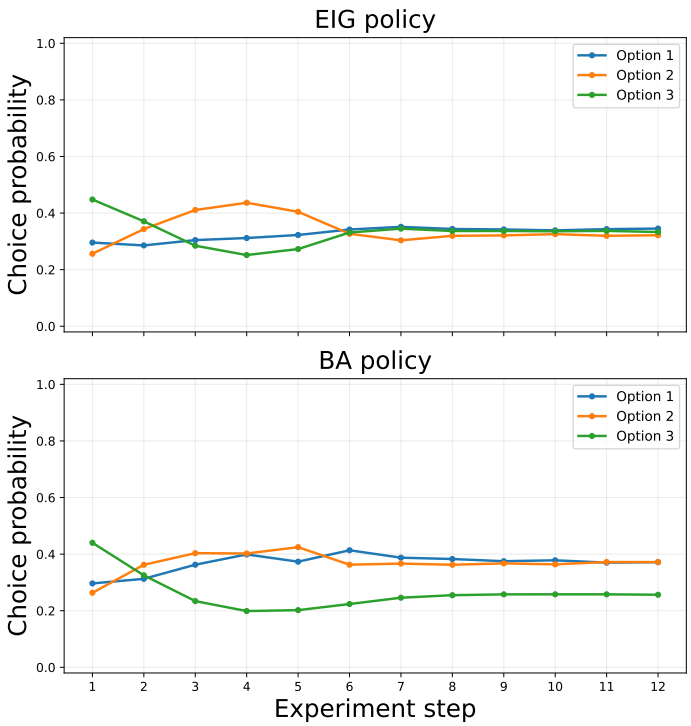


Figure 11: Average choice probabilities for the EIG and BA policies in the pizza choice example. For each parameter over a prior sample, evaluate the representative design traces and the corresponding choice probability at each experiment step. The averages of choice probabilities at each step are computed over the prior-sampled parameter values.

	Trained by EIG bound (Eq. (10))	Trained by BA bound (Eq. (33))
Freq. of domination of Option 1	0.1703	0.1395
Freq. of domination of Option 2	0.1598	0.1343
Freq. of domination of Option 3	0.1773	0.3052
Freq. of having at least one dominated option	0.4182	0.4800

Table 1: Frequency of attribute-wise dominated options in the pizza choice example, over prior-sampled parameter values and $T = 12$ experimental steps. Option j is counted as a dominated option when it is farther from the ideal point than another option on both spiciness and cheesiness. The last column gives the frequency with which at least one of the three options is dominated.

policies in terms of choice probabilities and choice contexts. As shown in Figure 11, the EIG policy generates designs in which the alternatives become progressively more balanced in their attribute profiles and the probability of being chosen over the course of the experiment. This pattern reflects its objective of maximizing EIG, as it is generally more informative when multiple choice outcomes remain plausible and share almost the same probability of being chosen. In contrast, the BA policy tends to maintain one relatively dominated alternative over experiment steps, which has lower chosen probability. It is worth noting that including an inferior option in the experimental design does not necessarily reduce its informativeness in this case. Although the dominated option is less likely to be selected, its presence activates the attraction-effect term $\gamma A_j(\boldsymbol{x}, \boldsymbol{\theta})$ to be non-zero in the utility function $V_j(\boldsymbol{x}, \boldsymbol{\theta})$, thereby providing additional information for estimating the associated parameters γ and λ. Conversely, if the experimental design does not deliberately include a dominated option, it may provide insufficient variation to identify the attraction-effect parameters, resulting in degenerate estimates such that $\hat{\gamma} = 0$ and $\hat{\lambda} \to +\infty$. Hence, the BA policy suggests a mechanism that may improve identification of these parameters. Table 1 further demonstrates the BA policy's advantages by showing that the frequency with which a dominated option appears across all tested experimental trajectories is 0.4800 under BA, compared with 0.4182 under EIG. As a result, EIG tends to generate relatively balanced alternatives, whereas BA deliberately includes a dominated alternative more frequently, achieving comparable mean information gain with lower variability in evidential performance.

6 Discussion

Our work has considered a design criterion based on the minimization of evidential bias, the bias against criterion, as an alternative to EIG. While there is extensive existing research on alternatives to EIG for batch design, they have been studied far less for amortized sequential design. Because the bias against is defined as a probability, directly optimizing it is difficult, and so our strategy has been to control the bias against by minimization of an upper bound based on the Paley-Zygmund inequality. Optimization of this bound can be formulated as maximization of a variance-penalized EIG criterion, which is both convenient to optimize and interpretable. Furthermore, the choice of penalty has a natural interpretation as corresponding to controlling the bias against at different evidential thresholds. Amortized sequential design has been implemented using the deep adaptive design approach of Foster et al. (2021), adapted to our new criterion. In several examples, including a complex discrete choice model, we are able to obtain designs which substantially improve on EIG-based designs in terms of the bias

against upper bound.

The main idea of our approach can be extended in several directions. In the discussion of bias against in Evans (2015, Section 4.6), a complementary concept is also defined, termed the bias in favour. While bias against is the probability of misleading evidence for the truth, the bias in favour is the probability that evidence is produced against a false value, with the false value chosen to differ from the truth by a practically significant amount. An obvious topic for future work is to combine the bias against with the bias in favour to develop design approaches that take into account which differences are practically significant to detect in a given problem.

Acknowledgements

David Nott's research was supported by the Ministry of Education, Singapore, under the Academic Research Fund Tier 2 (MOE-T2EP20123-0009).

A Proof of Proposition 1

We reproduce the proof for completeness. For $\delta \in [0,1]$, let $A = \{Z > \delta E(Z)\}$. Writing $\mathbb{I}_A$ and $\mathbb{I}_{A^c} = 1 - \mathbb{I}_A$ for the corresponding indicators,

$$\begin{aligned}(1-\delta)E(Z) &= E(Z - \delta E(Z)) = E((Z - \delta E(Z))\mathbb{I}_A) + E((Z - \delta E(Z))\mathbb{I}_{A^c}) \\ &\leq E((Z - \delta E(Z))\mathbb{I}_A),\end{aligned}$$

since $Z - \delta E(Z) \leq 0$ on A^c. By the Cauchy–Schwarz inequality,

$$E((Z - \delta E(Z))\mathbb{I}_A) \leq \sqrt{E((Z - \delta E(Z))^2)}\sqrt{P(A)}.$$

Combining these and using $E(Z) \geq 0$ (so that $(1-\delta)E(Z) \geq 0$ and the inequality can be squared without changing direction),

$$(1-\delta)^2 E(Z)^2 \leq E((Z - \delta E(Z))^2)\, P(A),$$

Observing that $E((Z - \delta E(Z))^2) = \mathrm{Var}(Z) + (1-\delta)^2 E(Z)^2$, it is sufficient to have $\mathrm{Var}(Z) > 0$ for this to be positive. Then rearranging the last inequality above gives (14). The complementary bound (15) follows since

$$P(A^c) = 1 - P(A) \leq 1 - \frac{(1-\delta)^2 E(Z)^2}{\mathrm{Var}(Z) + (1-\delta)^2 E(Z)^2} = \frac{\mathrm{Var}(Z)}{\mathrm{Var}(Z) + (1-\delta)^2 E(Z)^2}.$$

B The bias-against bound and variance-penalized EIG

Here we make precise the correspondence used in Section 3.2, namely that maximizing $R_t(x)$ is equivalent to maximizing the variance-penalized criterion at a suitable penalty. Throughout, write $\mu(x) := E(Z(y, x, \theta)) = \mathrm{EIG}(x)$ and $v(x) := \mathrm{Var}(Z(y, x, \theta))$ for a design x in some class $\mathcal{X}$, and for a fixed constant t define

$$R_t(x) := \frac{\mu(x) - t}{\sqrt{v(x)}}, \qquad L_\alpha(x) := \mu(x) - \alpha\, v(x).$$

The proof of Proposition 2 below is based on the quadratic transform for fractional programming (Shen and Yu, 2018, 2025), which for statisticians is naturally described using a minorization–maximization (MM) argument (Hunter and Lange, 2004). The minorization that is the basis of the proof derives from arguments in Shen and Yu (2025, Section VIII). We introduce some notation to simplify the discussion. For x with $v(x) > 0$ write $\zeta(x) := R_t(x)^2 = (\mu(x) - t)^2/v(x)$, and for a point z with $v(z) > 0$ define the surrogate

$$g(x \mid z) := 2\kappa(z)(\mu(x) - t) - \kappa(z)^2\, v(x), \qquad \kappa(z) := \frac{\mu(z) - t}{v(z)}. \tag{35}$$

Since

$$\zeta(x) - g(x \mid z) = \frac{\big[(\mu(x) - t) - \kappa(z)\, v(x)\big]^2}{v(x)} \;\geq\; 0, \tag{36}$$

with equality at $x = z$, the surrogate minorizes ζ and is tangent to it at z. An MM step for maximizing ζ is $z \leftarrow \arg\max_x g(x \mid z)$, and since $g(x \mid z) = 2\kappa(z)[L_{\kappa(z)/2}(x) - t]$ with $\kappa(z) > 0$ for $\mu(z) > t$, the MM step maximizes the variance-penalized criterion L_α at $\alpha = \kappa(z)/2$. Proposition 2 states that a maximizer of the ratio R_t is a fixed point of this iteration, with the surrogate there being L_{α^*}. No concavity of μ or convexity of v is required here, although that is often assumed in fractional programming problems. These assumptions are needed there to make the surrogate objective concave in the optimization steps and to guarantee convergence to a global maximizer. However, these assumptions are not necessary for the fixed-point property that Proposition 2 establishes.

To give some further intuition for Proposition 2 it can be helpful to consider what happens when μ and v are differentiable. In that case, at an interior maximizer of x^* of $R_t(x)$,

$$0 = \nabla_x R_t(x^*) = v(x^*)^{-1/2} \left(\nabla_x \mu(x^*) - \frac{\mu(x^*) - t}{2v(x^*)} \nabla_x v(x^*) \right),$$

and the term in brackets is the stationarity condition at x^* for the penalized criterion $L_\alpha(x)$ with $\alpha = \alpha^* = (\mu(x^*) - t)/(2v(x^*))$. The proof of Proposition 2 requires no differentiability assupmptions about μ and v.

Proposition 2. *Suppose $x^* \in \mathcal{X}$ maximizes R_t over $\{x \in \mathcal{X} : v(x) > 0\}$, and that $\mu(x^*) > t$ and $v(x^*) > 0$. Set*

$$\alpha^* = \frac{\mu(x^*) - t}{2\, v(x^*)}. \tag{37}$$

Then x^* maximizes L_{α^*} over $\{x \in \mathcal{X} : v(x) > 0\}$, and

$$\max_{x:\, v(x)>0} L_{\alpha^*}(x) = L_{\alpha^*}(x^*) = t + \frac{\mu(x^*) - t}{2} = \frac{\mu(x^*) + t}{2}. \tag{38}$$

Proof. Write $r^* := R_t(x^*) = (\mu(x^*) - t)/\sqrt{v(x^*)} > 0$ and $\kappa^* := \kappa(x^*) = (\mu(x^*) - t)/v(x^*) = 2\alpha^* > 0$, and note that

$$L_{\alpha^*}(x^*) = \mu(x^*) - \alpha^* v(x^*) = \mu(x^*) - \frac{\mu(x^*) - t}{2} = t + \frac{\mu(x^*) - t}{2}, \tag{39}$$

which is the same as (38). We show $L_{\alpha^*}(x) \le L_{\alpha^*}(x^*)$ for every x with $v(x) > 0$. There are two cases.

If $\mu(x) \le t$, then since $\alpha^* v(x) > 0$ (since $\mu(x^*) > t$ implies $\alpha^* > 0$, and $v(x) > 0$) we have

$$L_{\alpha^*}(x) = \mu(x) - \alpha^* v(x) < \mu(x) \le t < t + \frac{\mu(x^*) - t}{2} = L_{\alpha^*}(x^*),$$

using $\mu(x^*) > t$ and (39).

If $\mu(x) > t$, then $R_t(x) > 0$, and optimality of x^* gives $0 < R_t(x) \le r^*$. Squaring both sides of the last inequality,

$$\zeta(x) \le (r^*)^2 = \zeta(x^*). \tag{40}$$

Applying the minorization (36) at $z = x^*$, and using (40) and the tangency of the surrogate at x^*,

$$4\alpha^*\big[L_{\alpha^*}(x) - t\big] = g(x \mid x^*) \;\le\; \zeta(x) \;\le\; \zeta(x^*) = g(x^* \mid x^*) = 4\alpha^*\big[L_{\alpha^*}(x^*) - t\big],$$

where the first and last equalities use $g(x \mid x^*) = 2\kappa^*[L_{\kappa^*/2}(x) - t] = 4\alpha^*[L_{\alpha^*}(x) - t]$ and $g(x^* \mid x^*) = \zeta(x^*)$ follows from (36) at $x = z = x^*$. Dividing by $4\alpha^* > 0$ gives $L_{\alpha^*}(x) \le L_{\alpha^*}(x^*)$.

Combining the two cases, x^* maximizes L_{α^*} over $\{x \in \mathcal{X} : v(x) > 0\}$, with optimal value (39). □

Remark 1. Proposition 2 is stated for optimization over a design x, but its proof uses no structure of the set $\mathcal{X}$ beyond the values of the two functionals μ and v. The result therefore applies in the sequential adaptive setting of Section 4, with x replaced by the policy parameter φ, $\mathcal{X}$ by the policy parameter space, and $\mu(\varphi) = E\{Z(y, \varphi, \theta)\}$ and $v(\varphi) = \mathrm{Var}\{Z(y, \varphi, \theta)\}$ the mean and variance of the log relative belief ratio of the sequential experiment, provided a maximizer φ^* of R_t over the policy parameter space exists.

C Details of derivations in Example 1

C.1 Derivations of the inequalities (26) and (27) in Example 1

Here we derive (26) and (27) in the main text.

For (26), we first write $I = \frac{1}{2}\log(1+\tau^2/s^2)$, which is the EIG conditional on $\xi = 0$, and $\eta = y/(s^2+\tau^2)^{1/2}$. Using the expression (24) we can then write

$$P(K > 0|\xi = 1) = P\left(\frac{(y-\theta)^2}{s^2} < 2I + \eta^2 \;\middle|\; \xi = 1\right).$$

Given $\xi = 1$, y and θ are independent, with $\theta|\xi = 1 \sim N(0,\tau^2)$ and $y|\xi = 1 \sim N(0,\sigma_B^2)$, and so

$$P(K > 0|\xi = 1) = E_{y\sim p(y|\xi=1)}\left(P\left(\frac{(y-\theta)^2}{s^2} < 2I + \eta^2 \;\middle|\; y, \xi = 1\right)\right),$$

where $(y-\theta)/s|y,\xi = 1 \sim N(y/s, \tau^2/s^2)$, with this normal density being upper bounded by $s(2\pi\tau^2)^{-1/2}$. Multiplying the upper bound by the width of the interval $(-(2I+\eta^2)^{1/2}, (2I+\eta^2)^{1/2})$, we obtain a bound on the probability

$$\begin{aligned} P(K > 0|\xi = 1) &\le E_{y\sim p(y|\xi=1)}\left(2s(2\pi\tau^2)^{-1/2}(2I+\eta^2)^{1/2}\right) \\ &\le E_{y\sim p(y|\xi=1)}\left(2s(2\pi\tau^2)^{-1/2}((2I)^{1/2}+|\eta|)\right), \\ &\le Cs(\log n)^{1/2} \end{aligned}$$

for some constant C depending only on τ and σ_B, where in the second to last line we have used $(2I+\eta^2)^{1/2} \le ((2I)^{1/2}+|\eta|)$ and in the last line we have used $y|\xi = 1 \sim N(0,\sigma_B^2)$ and $I = 1/2\log n + O(1)$. Hence $P(K \le 0|\xi = 1) \ge 1 - Cs(\log n)^{1/2}$.

For (27), we have

$$P(K \le 0|\xi = 0) \le P\left(\log\left(1+\frac{\tau^2}{s^2}\right) \le W^2 \;\middle|\; \xi = 0\right),$$

where $W = (y-\theta)/s \sim N(0,1)$ when $\xi = 0$. Now $W^2|\xi = 0$ is χ_1^2 with moment generating function $M(t) = (1-2t)^{-1/2}$ for $|t| < 0.5$. Applying a Chernoff bound with $t = 1/4$ and noting that $M(1/4) = \sqrt{2}$ gives

$$P(K \le 0|\xi = 0) \le \sqrt{2}\left(1+\frac{\tau^2}{s^2}\right)^{-1/4}.$$

Noting that $\left(1+\frac{\tau^2}{s^2}\right)^{-1/2} = \left(\frac{s^2}{s^2+\tau^2}\right)^{1/2} \le \left(\frac{s^2}{\tau^2}\right)^{1/2} = s/\tau$, we see that

$$P(K \le 0|\xi = 0) \le \sqrt{2}(s/\tau)^{1/2},$$

which demonstrates (27).

C.2 Derivation of the inequality (28) in Example 1

We need some notation first. Let X, Y and Z be random variables. The mutual information of X and Y is

$$I(X;Y) := \mathrm{KL}(p(X,Y) \,\|\, p(X)p(Y)) \,.$$

The EIG is $I(\theta; y)$ which follows directly from (7). The conditional mutual information $I(X;Y|Z)$ is

$$I(X;Y|Z) := E_{Z\sim p(Z)} \left(\mathrm{KL}(p(X,Y|Z) \,\|\, P(X|Z)p(Y|Z))\right).$$

The chain rule for mutual information implies that

$$I(\theta; y, \xi) = I(\theta; y) + I(\theta; \xi|y),$$

which upon rearranging and using the property that mutual information and conditional mutual information are non-negative implies

$$I(\theta; y, \xi) - I(\theta; \xi|y) = I(\theta; y) = \mathrm{EIG} \leq I(\theta; y, \xi). \tag{41}$$

Now,

$$\begin{aligned} I(\theta; y, \xi) &= \mathrm{KL}(p(\theta, y, \xi) \,\|\, p(\theta)p(y, \xi)) \\ &= E_{\xi\sim p(\xi)} \left(\mathrm{KL}(p(\theta, y|\xi) \,\|\, p(\theta)p(y|\xi))\right) \\ &= (1-q)\mathrm{KL}(p(\theta, y|\xi=0) \,\|\, p(\theta)p(y|\xi=0)) + q\mathrm{KL}(p(\theta, y|\xi=1) \,\|\, p(\theta)p(y|\xi=1)) \,. \end{aligned} \tag{42}$$

Since θ and y are independent given $\xi = 1$, the second term in the last line above is zero. The first term in (42) is the conditional mutual information between θ and y given $\xi = 0$ (i.e. the EIG for the experiment where $\xi = 0$), the normal location problem, and this can be computed analytically using conjugacy so that we have

$$I(\theta; y, \xi) = \frac{1-q}{2} \log\left(1 + \frac{\tau^2}{s^2}\right). \tag{43}$$

Next, consider $I(\theta; \xi|y)$. We have

$$\begin{aligned} I(\theta; \xi|y) &= E_{y\sim p(y)} \left(\mathrm{KL}(p(\theta, \xi|y) \,\|\, p(\theta|y)p(\xi|y))\right) \\ &= E_{(y,\theta)\sim p(y,\theta)} \left(E_{\xi\sim p(\xi|y,\theta)}(\log p(\xi|y, \theta)\right) - E_{y\sim p(y)} \left(E_{\xi\sim p(\xi|y)}(\log p(\xi|y))\right) \end{aligned}$$

Recall from Section 2.3 the notation for the Shannon or differential entropy of a random variable X as $H(X) = -E_{X\sim p(X)}(\log p(X))$. If X is binary with $P(X = 1) = p$, then

$H(X) = -p \log p - (1-p)\log(1-p)$, and abusing notation we also write $H(p) := H(X)$. Then we have

$$\begin{aligned} I(\theta; \xi|y) &= E_{y\sim p(y)}(H(\xi|y)) - E_{(y,\theta)\sim p(y,\theta)}(H(\xi|y,\theta)) \\ &\leq E_{y\sim p(y)}(H(\xi|y)), \end{aligned} \tag{44}$$

where in the last line we have used the fact that Shannon entropy for a discrete variable is non-negative. The function $H(\cdot)$ is concave, so by Jensen's inequality

$$\begin{aligned} I(\theta; \xi|y) &\leq H(E_{y\sim p(y)}(P(\xi = 1|y)) \\ &= H(q). \end{aligned} \tag{45}$$

Hence from (41), (43) and (45) we obtain

$$\frac{1-q}{2}\log\left(1+\frac{\tau^2}{s^2}\right) - H(q) \leq \text{EIG} \leq \frac{1-q}{2}\log\left(1+\frac{\tau^2}{s^2}\right),$$

and recalling that $s^2 = \sigma^2/n$, we see that $\text{EIG} = \frac{1-q}{2}\log n + O(1)$. This demonstrates that $\text{EIG} \to \infty$.

D Differentiable relaxations for discrete examples

The hyperbolic-discounting and pizza choice examples both involve discrete simulated outcomes. This means that the reparametrization approach we have used to construct differentiable approximations to the design objective breaks down, since the simulated data is not a differentiable function of the policy parameters and noise. This occurs, for example, if the response is sampled as an exact Bernoulli or categorical variable. To obtain a differentiable approximation of the training objective, we use the so-called Gumbel-softmax trick (Jang et al., 2016; Maddison et al., 2016) to relax discrete distributions to continuous approximations.

An exact sample from a categorical distribution on $\{1, \ldots, K\}$ with respective probabilities $p_1, \ldots, p_K$ can be obtained by first drawing independent standard Gumbel variables

$$g_j = -\log\{-\log u_j\}, \qquad u_j \sim \text{Uniform}(0,1),$$

$j = 1, \ldots, K$, and then setting

$$y = \arg\max_{1\leq j\leq K}\{\log p_j + g_j\}.$$

Below we write $e_y = (e_{y1}, \ldots, e_{yK})$ for the one-hot encoding of y where $e_{yj} = 1$ if $y = j$ and $e_{yj} = 0$ otherwise.

The Gumbel-softmax relaxation of the distribution of e_y replaces the random variable e_{yj} with the continuous random variable

$$\widetilde{y}_j = \frac{\exp\{(\log p_j + g_j)/\tau\}}{\sum_{\ell=1}^{K} \exp\{(\log p_\ell + g_\ell)/\tau\}}, \qquad j = 1, \ldots, K.$$

where $\tau > 0$ is a temperature parameter. The distribution of $\widetilde{y} = (\widetilde{y}_1, \ldots, \widetilde{y}_K)$ approximates the distribution of e_y, with a draw of $\widetilde{y}$ becoming closer to a draw of e_y as $\tau \to 0$, and larger values of τ giving a smoother density for $\widetilde{y}$. In training, samples of $\widetilde{y}$ are used to approximate the design objective, and passed to the policy network input-history.

We use the Gumbel-softmax draw as a differentiable version of the simulated categorical response. The usual categorical log likelihood is

$$\log p_y = \sum_{j=1}^{K} e_{yj} \log p_j.$$

Since the continuous simulator replaces e_y by $\widetilde{y}$, the differentiable log-likelihood approximation is

$$\sum_{j=1}^{K} \widetilde{y}_j \log p_j.$$

In the hyperbolic-discounting example of subsection 5.2 the response is binary. For the binary response, let $\psi(x, \theta)$ denote the model probability of a 1 at design x. During training we replace the Bernoulli draw by

$$\widetilde{y} \in (0, 1),$$

where abusing notation $\widetilde{y}$ is what we have called $\widetilde{y}_2$ for the case of a categorical distribution on $\{0, 1\}$ with $K = 2$ in our previous discussion. The log likelihood term in the objective is approximated by

$$\widetilde{y} \log \psi(x, \theta) + (1 - \widetilde{y}) \log\{1 - \psi(x, \theta)\}.$$

Thus the simulator used in training is differentiable with respect to the design through $\psi(x, \theta)$, and the usual reparameterized EIG and BA objectives can be optimized by backpropagation.

In the pizza choice example of subsection 5.3, the response is a three-option categorical choice. Let

$$p_j(x, \theta) = P(y = j \mid x, \theta), \qquad j = 1, 2, 3,$$

be the choice probabilities for the three pizza options. During training we draw $\widetilde{y} = (\widetilde{y}_1, \widetilde{y}_2, \widetilde{y}_3)$ using

$$\widetilde{y}_j = \frac{\exp\{(\log p_j(x, \theta) + g_j)/\tau\}}{\sum_{\ell=1}^{3} \exp\{(\log p_\ell(x, \theta) + g_\ell)/\tau\}}.$$

The log-likelihood contribution is then approximated by

$$\sum_{j=1}^{3} \widetilde{y}_j \log p_j(x, \theta).$$

which is the differentiable analogue of the usual categorical log likelihood.

The Gumbel-softmax relaxation is used only for training the policies. For final design-trace evaluation and interpretation of the learned policies, we return to the intended discrete distributions, using Bernoulli responses in the hyperbolic example and discrete categorical responses in the pizza choice example. For the hyperbolic-discounting example and pizza choice examples, we use $\tau = 0.2$ and 0.5 respectively in the Gumbel-softmax distribution.

Especially for the pizza case study, two additional steps for smoothness and differentiability are required. First, for policy net training, the data simulator uses a soft domination function $D_{jk}^{\text{soft}}(x, \theta)$ to ensure differentiability and smoothness. Specifically, an independent standard sigmoid function $\sigma(.)$ is applied on each attribute dimension such that

$$D_{jk}^{\text{soft}}(x, \theta) = \prod_{a=1}^{2} \sigma\big(c\{(x_{ka} - \mu_a)^2 - (x_{ja} - \mu_a)^2\}\big)\,,$$

where the scale parameter $c > 0$ controls the sharpness with which the dominance between the compared options is perceived. The soft dominance function $D_{jk}^{\text{soft}}(x, \theta)$ constructs a smooth and differentiable data simulator for BOED, which converges to the strict dominance indicator as $c \to \infty$. Therefore, we fix a sufficiently large c, $c = 30$, in the discrete choice experiment.

Lastly, after policy net training, differentiability is also required in the mapping from the policy-network output to the design space. We defined the design space as the closed box $[0, 1]^6$, while 0 means no spiciness or cheesiness and 1 means extreme spiciness or cheesiness. If this constraint is imposed by clipping the raw policy output, then any coordinate clipped to 0 or 1 has zero local gradient through the clipping operation. To avoid this during training, the policy first outputs an unconstrained raw vector

$$z_t = \eta_\varphi(\mathcal{I}_{t-1}) \in \mathbb{R}^6.$$

This vector is arranged in a centre-and-offset form,

$$\widetilde{x}_t = (z_{t1}, z_{t2}, z_{t1} + r_{\text{loc}} z_{t3}, z_{t2} + r_{\text{loc}} z_{t4}, z_{t1} + r_{\text{loc}} z_{t5}, z_{t2} + r_{\text{loc}} z_{t6})\,,$$

where $r_{\text{loc}} = 0.65$ controls the spread of the second and third pizza options around the first. The training design is then

$$x_t = \sigma(\widetilde{x}_t / \tau_x) \in (0, 1)^6,$$

where $\tau_x = 3$. This gives a smooth differentiable map from $\eta_\varphi(\mathcal{I}_{t-1})$ to the design used in the relaxed simulator. The centre-and-offset form is a policy parametrization, not a change to the choice model. For design-trace evaluation, we round the resulting continuous designs x_t to a grid with spacing 0.01 so that the reported attributes lie on interpretable levels in the original closed design space.

E Training and Evaluation procedure

In this section we describe the training procedure, evaluation procedure, and hyperparameters used for the three examples in the main manuscript. For each example, the goal is to learn a policy network η_φ, where φ denotes the trainable network parameters. We use the Adam optimizer with initial learning rate lr, and write β for the two Adam coefficients used to compute running averages of the gradient and its square. Similar to Foster et al. (2021), we use an ExponentialLR scheduler with multiplicative learning-rate decay factor γ.

For the policy network, we use the set-equivariant architecture of Foster et al. (2021). At step t, the network takes the previous history $\{(x_s, y_s) : s < t\}$ as input. Each past design-outcome pair is first mapped to an embedding of dimension d_e by an encoder network with h_e hidden layers and hidden width d_h. The embeddings are then pooled over the history and passed to an emitter network with h_m hidden layers and hidden width d_h, which outputs the next design.

For the source-location example, EIG and BA policies are trained from random initialization for N_{train} epochs. At each epoch, a Monte Carlo estimate of the relevant objective is formed using batch size $\mathcal{B}$, contrastive sample size L, and outer Monte Carlo sample size J. The Monte Carlo variables entering the training objective are resampled at every epoch.

For the hyperbolic-discounting and pizza choice examples, we use a common warm-start and fine-tune training strategy. We first train a warm start policy net with objective (10) for $N_{warm} = 8000$ epochs. This policy is then copied and used as the initialization for an additional fine-tune $N_{cont} = 2000$ epochs of objective-specific training. One copy continues to optimize the EIG lower-bound objective (10); this gives the reported EIG policy. The other copies optimize the BA objective (33) for different values of α; these give the reported BA policies. Thus, in these examples, EIG and BA policies are compared after the same warm start and the same number of fine-tune epochs, so that the difference is only the training objective in fine-tuning. Similar warm-start and fine-tuning strategies are widely used in neural-network optimization (Erhan et al., 2010). In both the warm-up and fine-tuning stages, we use windowed model checkpointing with window length l_w: within each window of l_w

epochs, we store the model parameters that attain the lowest training loss, and at the end of the window we reload these parameters before continuing training.

After training, we evaluate the learned policies using batch size $\mathcal{B}_{eval}$, contrastive sample size L_{eval}, and outer Monte Carlo sample size J_{eval}. These evaluations give the estimates $\overline{Z}(\varphi)$, $\sqrt{S(\varphi)}$, and

$$\widehat{R}(\varphi) = \frac{\overline{Z}(\varphi)}{\sqrt{S(\varphi)}},$$

which are shown in Figures 1, 4, and 8. For the design-trace figures, for each fixed parameter value we run the policy tr_{repeat} times and obtain a sample of stochastic design traces. We treat each trace as an ordered sequence. If $X^{(m)} = (x_1^{(m)}, \ldots, x_T^{(m)})$ denotes the mth trace, we stack it as $v^{(m)} = \mathrm{vec}(X^{(m)})$ and summarize the repeated traces by the geometric median

$$\widehat{v} = \arg\min_{v} \sum_{m=1}^{tr_{repeat}} \|v - v^{(m)}\|_2.$$

The vector $\widehat{v}$ is reshaped to obtain the representative design trace, and the geometric median is computed by Weiszfeld iteration (Weiszfeld, 1937) for tr_{iter} iterations. We repeat this over a prior sample of size tr_{sample}, compute the DTW distance between the representative EIG and BA traces (Giorgino, 2009), and display traces for selected distance quantiles together with manually chosen parameter values.

For the pizza choice example, the geometric median is computed separately for each of the three option trajectories, and the policy distance is the sum of the DTW distances over the three option trajectories.

E.1 Monte Carlo robustness check for the pizza-choice example

A natural concern is whether the results are sensitive to the choice of hyperparameters. To assess whether the comparison between the EIG and BA policies is sensitive to Monte Carlo sample sizes, we repeated the pizza-choice objective evaluation for several combinations of the evaluation batch size $\mathcal{B}_{\mathrm{eval}}$, contrastive sample size L_{eval}, and outer Monte Carlo sample size J_{eval}. Figure 12 shows that the qualitative comparison is stable across these settings: the two policies achieve similar values of $\overline{Z}$, while the BA policy generally has smaller $\sqrt{S}$ and consequently larger $\overline{Z}/\sqrt{S}$. The comparison in Figure 8 therefore does not appear to be driven by the finite-sample Monte Carlo approximation.

Hyperparameter	Value
Training epochs N_{train}	10000
Experiment steps T	30
Batch size $\mathcal{B}$	5000
Contrastive sample size L	100
Outer Monte Carlo sample size J	5000
Learning rate lr	5×10^{-5}
Adam coefficients β	$(0.8, 0.998)$
Scheduler decay factor γ	0.98
Gradient clipping	max norm 1.0
Evaluation batch size $\mathcal{B}_{eval}$	5000
Evaluation contrastive size L_{eval}	100
Evaluation outer size J_{eval}	5000
Trace prior sample size tr_{sample}	500
Trace repeats tr_{repeat}	30
Geometric-median iterations tr_{iter}	100
Embedding dimension d_e	16
Hidden width d_h	128
Encoder hidden depth h_e	1
Emitter hidden depth h_m	0

Table 2: Hyperparameter settings used for the source-location example.

Hyperparameter	Value
Training epochs N_{warm}	8000
Fine-tuning epochs N_{cont}	2000
Experiment steps T	20
Batch size $\mathcal{B}$	2000
Contrastive sample size L	1000
Outer Monte Carlo sample size J	400
Learning rate lr	5×10^{-6}
Adam coefficients β	$(0.9, 0.999)$
Scheduler decay factor γ	0.98
Window length l_w	500
Evaluation batch size $\mathcal{B}_{eval}$	2000
Evaluation contrastive size L_{eval}	5000
Evaluation outer size J_{eval}	2000
Trace prior sample size tr_{sample}	500
Trace repeats tr_{repeat}	30
Geometric-median iterations tr_{iter}	100
Embedding dimension d_e	16
Hidden width d_h	256
Encoder hidden depth h_e	2
Emitter hidden depth h_m	2

Table 3: Hyperparameter settings used for the hyperbolic-discounting example.

Hyperparameter	Value
Training epochs N_{warm}	8000
Fine-tuning epochs N_{cont}	2000
Experiment steps T	12
Batch size $\mathcal{B}$	500
Contrastive sample size L	500
Outer Monte Carlo sample size J	500
Learning rate lr	5×10^{-5}
Adam coefficients β	$(0.8, 0.998)$
Scheduler decay factor γ	0.98
Window length l_w	1000
Evaluation batch size $\mathcal{B}_{eval}$	2000
Evaluation contrastive size L_{eval}	1000
Evaluation outer size J_{eval}	4000
Trace prior sample size tr_{sample}	500
Trace repeats tr_{repeat}	30
Geometric-median iterations tr_{iter}	100
Embedding dimension d_e	64
Hidden width d_h	256
Encoder hidden depth h_e	2
Emitter hidden depth h_m	2

Table 4: Hyperparameter settings used for the pizza choice example.

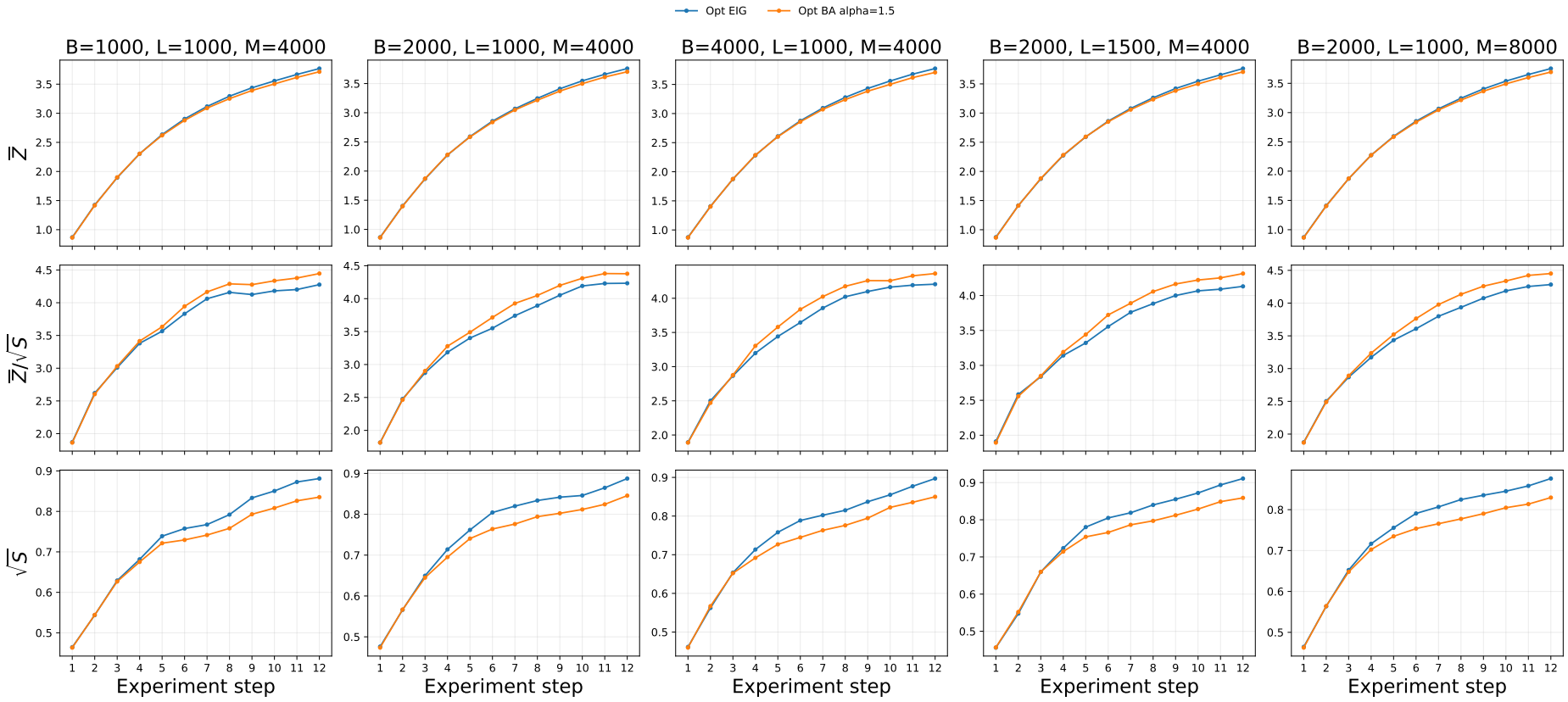


Figure 12: Monte Carlo robustness check for the pizza-choice example, comparing the EIG policy with the selected BA policy ($\alpha = 1.5$). From top to bottom, the rows show $\overline{Z}$, $\overline{Z}/\sqrt{S}$, and $\sqrt{S}$ over the $T = 12$ experimental steps. The columns correspond to different combinations of $(\mathcal{B}_{\text{eval}}, L_{\text{eval}}, J_{\text{eval}})$, as indicated above the panels. The qualitative comparison between the policies remains stable across the evaluation settings.

References

Barber, D. and Agakov, F. V. (2003). The IM algorithm: A variational approach to information maximization. In *Advances in Neural Information Processing Systems 16*, pages 201–208, Cambridge, MA. MIT Press.

Blau, T., Bonilla, E. V., Chades, I., and Dezfouli, A. (2022). Optimizing sequential experimental design with deep reinforcement learning. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 2107–2128. PMLR.

Blau, T., Chades, I., Dezfouli, A., Steinberg, D., and Bonilla, E. V. (2023). Statistically efficient Bayesian sequential experiment design via reinforcement learning with cross-entropy estimators. *arXiv preprint arXiv:2305.18435*.

Blume, J. D. (2002). Likelihood methods for measuring statistical evidence. *Statistics in Medicine*, 21(17):2563–2599.

Bracher, N., Kühmichel, L., Ivanova, D. R., Intes, X., Bürkner, P.-C., and Radev, S. T. (2026). JADAI: Jointly amortizing adaptive design and Bayesian inference. In *Proceedings of the 43rd International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR. arXiv:2512.22999.

Chaloner, K. and Verdinelli, I. (1995). Bayesian experimental design: A review. *Statistical Science*, 10(3):273 – 304.

Chen, A., Nott, D. J., and Tan, L. S. L. (2025). Weighted Fisher divergence for high-dimensional Gaussian variational inference. *arXiv preprint arXiv:2503.04246*.

Clarke, B. S. and Barron, A. R. (1990). Information-theoretic asymptotics of Bayes methods. *IEEE Transactions on Information Theory*, 36(3):453–471.

Cooper, L. G. and Nakanishi, M. (1983). Two logit models for external analysis of preferences. *Psychometrika*, 48(4):607–620.

Dinkelbach, W. (1967). On nonlinear fractional programming. *Management Science*, 13(7):492–498.

Erhan, D., Bengio, Y., Courville, A., Manzagol, P.-A., Vincent, P., and Bengio, S. (2010). Why does unsupervised pre-training help deep learning? *Journal of machine learning research*, 11(2).

Evans, M. (2015). *Measuring Statistical Evidence Using Relative Belief*. Taylor & Francis.

Foster, A., Ivanova, D. R., Malik, I., and Rainforth, T. (2021). Deep adaptive design: Amortizing sequential Bayesian experimental design. In Meila, M. and Zhang, T., editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 3384–3395.

Foster, A., Jankowiak, M., O'Meara, M., Teh, Y. W., and Rainforth, T. (2020). A unified stochastic gradient approach to designing Bayesian-optimal experiments. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 2959–2969. PMLR.

Giorgino, T. (2009). Computing and visualizing dynamic time warping alignments in R: The dtw package. *Journal of Statistical Software*, 31(7):1–24.

Huan, X., Jagalur, J., and Marzouk, Y. (2024). Optimal experimental design: Formulations and computations. *Acta Numerica*, 33:715–840.

Huan, X. and Marzouk, Y. M. (2016). Sequential Bayesian optimal experimental design via approximate dynamic programming. *arXiv preprint arXiv:1604.08320*.

Huang, D., Guo, Y., Acerbi, L., and Kaski, S. (2024). Amortized Bayesian experimental design for decision-making. In Globerson, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J., and Zhang, C., editors, *Advances in Neural Information Processing Systems*, volume 37, pages 109460–109486. Curran Associates, Inc.

Huang, D., Wen, X., Bharti, A., Kaski, S., and Acerbi, L. (2026). ALINE: Joint amortization for Bayesian inference and active data acquisition. *Advances in Neural Information Processing Systems*, 38:54068–54102.

Hunter, D. R. and Lange, K. (2004). A tutorial on MM algorithms. *The American Statistician*, 58(1):30–37.

Ivanova, D. R., Foster, A., Kleinegesse, S., Gutmann, M. U., and Rainforth, T. (2021). Implicit deep adaptive design: Policy-based experimental design without likelihoods. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in*

Neural Information Processing Systems, volume 34, pages 25785–25798. Curran Associates, Inc.

Ivanova, D. R., Hedman, M., Guan, C., and Rainforth, T. (2024). Step-DAD: Semi-amortized policy-based Bayesian experimental design. In *ICLR 2024 Workshop on Data-centric Machine Learning Research (DMLR): Harnessing Momentum for Science*.

Jang, E., Gu, S., and Poole, B. (2016). Categorical reparameterization with Gumbel-softmax. *arXiv preprint arXiv:1611.01144*.

Kingma, D. P. and Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.

Kullback, S. and Leibler, R. A. (1951). On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79 – 86.

Lim, V., Novoseller, E., Ichnowski, J., Huang, H., and Goldberg, K. (2022). Policy-based Bayesian experimental design for non-differentiable implicit models. *arXiv preprint arXiv:2203.04272*.

Lindley, D. V. (1956). On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics*, 27(4):986 – 1005.

Long, Q., Scavino, M., Tempone, R., and Wang, S. (2013). Fast estimation of expected information gains for Bayesian experimental designs based on Laplace approximations. *Computer Methods in Applied Mechanics and Engineering*, 259:24–39.

Maddison, C. J., Mnih, A., and Teh, Y. W. (2016). The concrete distribution: A continuous relaxation of discrete random variables. *arXiv preprint arXiv:1611.00712*.

Paley, R. E. A. C. and Zygmund, A. (1932). A note on analytic functions in the unit circle. *Mathematical Proceedings of the Cambridge Philosophical Society*, 28(3):266–272.

Phillips, A., Kerrigan, G., and Rainforth, T. (2026). Bayesian experimental design via score matching. *arXiv preprint arXiv:2607.08335*.

Rainforth, T., Cornish, R., Yang, H., Warrington, A., and Wood, F. (2018). On nesting Monte Carlo estimators. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 4267–4276. PMLR.

Rainforth, T., Foster, A., Ivanova, D. R., and Smith, F. B. (2024). Modern Bayesian experimental design. *Statistical Science*, 39(1):100 – 114.

Rockafellar, R. T. and Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–41.

Rooderkerk, R. P., Van Heerde, H. J., and Bijmolt, T. H. (2011). Incorporating context effects into a choice model. *Journal of Marketing Research*, 48(4):767–780.

Rossa, T., Phillips, A., and Rainforth, T. (2026). Action-BED: Task-driven Bayesian experimental design with singly intractable objectives. *arXiv preprint arXiv:2606.23662*.

Royall, R. (2000). On the probability of observing misleading statistical evidence (with discussion). *Journal of the American Statistical Association*, 95(451):760–768.

Royall, R. M. (1997). *Statistical Evidence: A Likelihood Paradigm*, volume 71 of *Monographs on Statistics and Applied Probability*. Chapman & Hall, London.

Ryan, E. G., Drovandi, C. C., McGree, J. M., and Pettitt, A. N. (2016). A review of modern computational algorithms for Bayesian optimal design. *International Statistical Review*, 84(1):128–154.

Shen, K. and Yu, W. (2018). Fractional programming for communication systems—Part I: Power control and beamforming. *IEEE Transactions on Signal Processing*, 66(10):2616–2630.

Shen, K. and Yu, W. (2025). Quadratic transform for fractional programming in signal processing and machine learning. *IEEE Signal Processing Magazine*. arXiv:2503.09977.

Shen, W. (2023). *Reinforcement learning based sequential and robust Bayesian optimal experimental design*. PhD thesis, University of Michigan.

Shen, W., Dong, J., and Huan, X. (2025). Variational sequential optimal experimental design using reinforcement learning. *Computer Methods in Applied Mechanics and Engineering*, 444:118068.

Shen, W. and Huan, X. (2023). Bayesian sequential optimal experimental design for nonlinear models using policy gradient reinforcement learning. *Computer Methods in Applied Mechanics and Engineering*, 416:116304.

Shen, W. and Huan, X. (2026). Mean-variance risk-aware Bayesian optimal experimental design for nonlinear models. *arXiv preprint arXiv:2604.04315*.

Sheng, X. and Hu, Y.-H. (2005). Maximum likelihood multiple-source localization using acoustic energy measurements with wireless sensor networks. *IEEE Transactions on Signal Processing*, 53(1):44–53.

Weiszfeld, E. (1937). Sur le point pour lequel la somme des distances de n points donnés est minimum. *Tohoku Mathematical Journal*, 43:355–386.

Xu, A. and Davenport, M. (2020). Simultaneous preference and metric learning from paired comparisons. *Advances in Neural Information Processing Systems*, 33:454–465.